INSIGHTSIAS
SIMPLIFYING IAS EXAM PREPARATION

Secure Synopsis compilation for April-2026

General Studies-1

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Indian culture will cover the salient aspects of Art Forms, literature and Architecture from ancient to modern times.

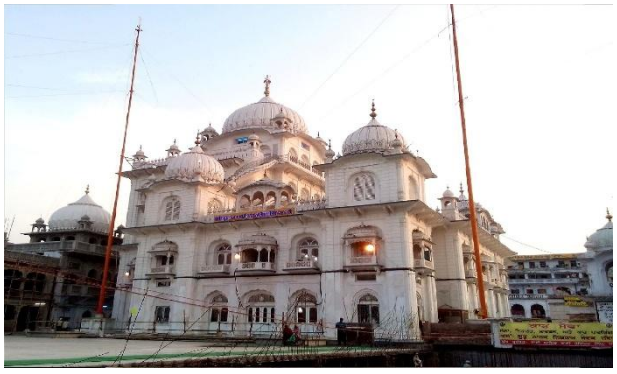
Q. The architectural evolution of Sikh shrines reflects both indigenous traditions and adaptive responses to historical challenges. Elucidate. (10 M)

Introduction

Emerging from a socio-religious movement rooted in egalitarianism, Sikh sacred spaces evolved from modest congregational structures to architecturally refined complexes. Their form reflects a dynamic interplay between **indigenous building traditions** and **historical contingencies**.

Body





<u>Indigenous traditions in Sikh shrine architecture</u>

1. **Use of local building idioms and materials:** Early Gurudwaras adopted **vernacular construction styles** of Punjab such as brick masonry and lime plaster, ensuring climatic suitability and cultural

continuity.
Eg: The early structure of **Harmandir Sahib (16th century, Guru Arjan Dev, 1588–1604)** used **brick and lime mortar**, typical of regional architecture, reflecting local craftsmanship.

2. **Sarovar-centric spatial planning:** Centrality of the **sacred water tank (sarovar)** reflects Indic traditions of ritual purification integrated with spatial design.
   Eg: The **Amrit Sarovar** around Harmandir Sahib symbolises **spiritual cleansing and equality**, with devotees accessing it irrespective of social status.
3. **Egalitarian layout and multiple entrances:** Sikh philosophy of equality is embedded through **four entrances** symbolising openness to all directions and communities.
   Eg: Harmandir Sahib's **four gateways** represent **universal accessibility**, aligned with Sikh doctrine of **sarbat da bhala**.
4. **Integration of congregational spaces:** Emphasis on **sangat (collective worship)** led to large open halls rather than hierarchical temple layouts.
   Eg: The **Darbar Sahib hall** facilitates **collective prayer and kirtan**, unlike garbhagriha-centric Hindu temples.
5. **Minimal iconography and emphasis on scripture:** Absence of idol worship influenced simpler interiors focused on **Guru Granth Sahib** as the spiritual centre.
   Eg: The sanctum houses the **Guru Granth Sahib**, reinforcing Sikhism's **text-centric devotional practice**.

<u>Adaptive responses to historical challenges</u>

1. **Incorporation of Mughal architectural elements:** Interaction with Mughal period led to adoption of **domes, arches, and pietra dura-like ornamentation**.
   Eg: The **gold-plated dome (early 19th century, patronage of Maharaja Ranjit Singh)** shows **Mughal influence blended with Sikh aesthetics**.
2. **Fortified religious architecture during conflict periods:** Turbulent 18th century led to construction of **defensive features** in some shrines.
   Eg: Akal Takht (est. 1606 by Guru Hargobind) symbolised **temporal authority (Miri-Piri)** and also functioned as a centre of resistance.
3. **Reconstruction and restoration after damage:** Repeated destruction during invasions led to cycles of rebuilding, influencing structural modifications.
   Eg: Harmandir Sahib was **rebuilt multiple times in the 18th century after Afghan invasions**, leading to reinforced structures.
4. **Monumentalisation under Sikh Empire:** Political consolidation under **Maharaja Ranjit Singh (early 19th century)** enabled large-scale embellishment and architectural grandeur.
   Eg: Extensive **gilding and marble inlay work** at Harmandir Sahib enhanced its **monumental character**.
5. **Modern conservation and expansion practices:** Contemporary challenges have led to incorporation of **modern materials and conservation techniques** while retaining core layout.
   Eg: Ongoing restoration and management by **Shiromani Gurdwara Parbandhak Committee (SGPC)** ensures preservation alongside **modern crowd management infrastructure**.

Conclusion

Sikh shrine architecture stands as a living testament to continuity anchored in indigenous ethos and

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resilience shaped by historical realities. Its evolution reflects how faith traditions creatively negotiate change while preserving core identity.

Q. Trace the evolution of manuscript traditions in India. Analyse their role in preserving literary heritage. (15 M)

Introduction

India's civilisational memory has travelled not through monuments alone but through manuscripts that encoded knowledge across generations. These fragile yet enduring carriers of thought have sustained India's literary continuum from antiquity to the digital age.

Body

Evolution of manuscript traditions in India

1. **Oral to written transition:** Early Vedic knowledge was preserved orally with precision before gradually being codified into manuscripts.
   Eg: Rigveda and other Vedic texts were transmitted orally through **shruti tradition** and later written on **birch bark and palm leaves** (standard historiography).
2. **Material diversity and regional variation:** Manuscripts evolved using locally available materials like palm leaves in the south and birch bark in the north.
   Eg: Bakhshali manuscript (birch bark, c. 3rd–7th century CE) reflects early mathematical texts in north-western India (Oxford studies).
3. **Institutionalisation in ancient centres:** Universities and monasteries became hubs for manuscript production, preservation and copying.
   Eg: Nalanda and Vikramashila universities (5th–12th century CE) housed extensive manuscript libraries (Xuanzang accounts).
4. **Temple and court patronage in medieval period:** Manuscript creation expanded under royal and religious patronage with illustrated texts.
   Eg: Jain Kalpasutra manuscripts (14th–15th century) with miniature paintings show artistic-literary fusion (Western Indian tradition).
5. **Perso-Arabic and multilingual expansion:** Indo-Islamic period saw growth of Persian manuscripts and translation movements.
   Eg: Akbar's Maktab Khana (16th century) translated Sanskrit texts like **Mahabharata into Persian (Razmnama)**.
6. **Colonial collection and archival phase:** European scholars collected, catalogued and preserved Indian manuscripts in global institutions.
   Eg: Shikshapatri manuscript (1826) presented to **Sir John Malcolm in 1830**, later preserved in **Bodleian Library, Oxford**.
7. **Modern digitisation and conservation efforts:** Post-independence India has focused on scientific preservation and digital archiving.
   Eg: National Mission for Manuscripts (launched 2003, Ministry of Culture) has documented over **50 lakh manuscripts** (official data).

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Role of manuscript traditions in preserving literary heritage

1. **Transmission of classical knowledge:** Manuscripts preserved foundational texts across philosophy, science, literature and religion.
   Eg: Charaka Samhita and Sushruta Samhita manuscripts ensured continuity of **Ayurvedic knowledge** (traditional medical corpus).
2. **Safeguarding linguistic diversity:** Manuscripts exist in multiple scripts and languages, reflecting India's plural literary culture.
   Eg: Sanskrit, Pali, Prakrit, Tamil and Persian manuscripts preserved in archives like **Asiatic Society, Kolkata**.
3. **Continuity of religious and ethical traditions:** Sacred texts guided social conduct and spiritual life across generations.
   Eg: Shikshapatri (212 Sanskrit shlokas) continues to guide **ethical living among Swaminarayan followers globally**.
4. **Preservation of regional histories and knowledge systems:** Local chronicles and texts captured socio-cultural and political histories.
   Eg: Bakhar literature in Maharashtra records regional history and governance practices (Maratha period sources).
5. **Artistic and aesthetic heritage:** Illustrated manuscripts contributed to development of painting traditions and visual culture.
   Eg: Mughal and Rajput miniature manuscripts combined literature with **fine arts traditions**.
6. **Source for modern research and historiography:** Manuscripts serve as primary sources for reconstructing India's past.
   Eg: Epigraphic and manuscript studies by ASI and Indologists have shaped understanding of ancient India (standard academic work).
7. **Cultural diplomacy and global recognition:** Manuscripts preserved abroad enhance global appreciation of Indian heritage.
   Eg: Bodleian Library's 2026 Shikshapatri bicentenary tour showcases India's literary heritage to global audiences.

Conclusion

Manuscript traditions embody India's intellectual resilience, preserving diversity across time and space. Their future lies in combining conservation with accessibility, ensuring that this living heritage continues to inspire both scholarship and society.

Q. "National symbols in colonial India were instruments of political mobilisation as well as ideological contestation. Examine this statement. Discuss its impact on the freedom struggle. (10 M)

Introduction

In colonial India, nationalism evolved through cultural idioms where symbols translated abstract political ideas into mass consciousness. These symbols became powerful instruments to mobilise people while also reflecting competing visions of Indian identity.

Body

National symbols as instruments of political mobilisation

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1. **Emotional unification of masses:** Symbols created a shared identity across caste, class, and region, enabling collective mobilisation.
   Eg: "Vande Mataram" (1882) became a rallying slogan during the **Swadeshi Movement (1905)**, uniting people against the **Partition of Bengal**.
2. **Simplification of nationalist message:** Symbols communicated complex ideas like swaraj in simple, relatable forms for the masses.
   Eg: Charkha popularised by Gandhi (1920s) symbolised **self-reliance and boycott of British goods**, making nationalism accessible.
3. **Visual assertion of resistance:** Flags and symbolic acts provided visible expressions of defiance against colonial authority.
   Eg: Tricolour flag (1931 adoption) was widely used in protests as a symbol of **national unity and resistance**.
4. **Mobilisation through cultural platforms:** Traditional and religious symbols were used to politicise society.
   Eg: Ganapati and Shivaji festivals (1890s) by Tilak transformed cultural events into **platforms for political awakening**.

National symbols as sites of ideological contestation

1. **Divergent community interpretations:** Different communities perceived symbols differently, reflecting plural identities.
   Eg: "Vande Mataram" faced opposition from sections of Muslims due to its **religious connotations**, leading to debates within Congress.
2. **Tension between inclusive and cultural nationalism:** Symbols often reflected competing visions of nationhood.
   Eg: Congress leadership moderated use of certain symbols in 1930s–40s to maintain **inclusive nationalism**.
3. **Regional and linguistic sensitivities:** Symbols did not uniformly resonate across diverse regions.
   Eg: Opposition to **Hindi-centric nationalism in South India** highlighted linguistic contestation.
4. **Ideological debates within nationalist leadership:** Leaders differed on use and meaning of symbols.
   Eg: Differences between **extremists and moderates** over use of assertive cultural symbols shaped political strategies.

Impact on the freedom struggle

1. **Transformation into mass movement:** Symbols helped convert elite nationalism into a broad-based popular struggle.
   Eg: Non-Cooperation Movement (1920–22) saw mass participation through **khadi adoption and symbolic boycott**.
2. **Strengthening national consciousness:** Symbols fostered unity and collective identity among diverse groups.
   Eg: Flag hoisting and national songs became acts of **everyday resistance** across India.
3. **Exposure of internal diversities:** Contestations highlighted plurality within Indian nationalism.
   Eg: Debates over symbols led to **greater accommodation of diverse identities under Gandhi's leadership**.
4. **Shaping inclusive political strategies:** Leadership adapted symbols to avoid alienation and sustain unity.
   Eg: Selective use of symbols ensured **broad-based participation across communities** in later phases of movement.

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Conclusion
National symbols acted as both unifying forces and arenas of debate, shaping a dynamic and inclusive nationalism. Their dual role ultimately strengthened the resilience and adaptability of the Indian freedom struggle.

Q. Examine the role of technological adaptation in Babur's success. How did it differ from earlier Indian warfare practices? (10 M)

Introduction
The early 16th century marked a critical phase where warfare began transitioning from traditional mass-based combat to technique-driven engagements. Babur's success illustrates how adaptation of existing technologies can decisively alter historical outcomes.

Body

Role of technological adaptation in Babur's success

1. **Integration of firearms with field tactics:** Babur effectively deployed matchlocks and artillery in open-field battles rather than restricting them to sieges.
   Eg: At the **First Battle of Panipat (1526)**, Babur used **field artillery with matchlockmen protected by mantlets**, enabling continuous and coordinated fire against advancing Lodi forces.
2. **Innovative use of Tulughma formation:** He combined mobility with discipline through flanking tactics and encirclement strategies.
   Eg: Babur's use of **Tulughma (division of army into flanking units)** allowed swift manoeuvring and encirclement, breaking the rigid formation of Ibrahim Lodi's army.
3. **Defensive-offensive cart system:** Use of carts tied with rawhide created a mobile barricade protecting gunners while enabling offensive action.
   Eg: At Panipat, **araba (cart) system** formed a defensive line with gaps for cavalry charges, combining static defence with dynamic attack.
4. **Coordination between cavalry and infantry:** Babur ensured synergy between traditional cavalry strength and new infantry-based gunpowder units.
   Eg: The **simultaneous use of cavalry charges with artillery fire** disrupted enemy formations and prevented regrouping.
5. **Adaptation of Central Asian techniques to Indian conditions:** Babur modified tactics learned from Ottoman influences to suit Indian battlefield realities.
   Eg: Adoption of **Ottoman-style gunpowder tactics after Battle of Chaldiran (1514)**, but adapted for Indian plains and dense enemy formations.

How it differed from earlier Indian warfare practices

1. **Shift from siege-centric to field use of artillery:** Earlier Indian powers largely confined artillery to fortifications and sieges.
   Eg: Pre-1526 evidence from **Bahmani and Gujarat kingdoms** shows artillery mainly used in sieges, not decisive field battles.
2. **Decline of elephant-centric warfare:** Traditional Indian armies relied heavily on war elephants, limiting mobility and tactical flexibility.
   Eg: Ibrahim Lodi's army at Panipat used **large war elephants**, which became ineffective against coordinated gunpowder attacks.
3. **From mass warfare to strategic manoeuvring:** Earlier battles emphasized numerical strength rather than tactical precision.
   Eg: Lodi forces were **numerically superior (over 50,000 troops)** but lacked coordinated formations, leading to battlefield congestion.

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4. **Limited integration of infantry firearms:** Infantry with firearms existed but lacked organisation and battlefield role.
   Eg: Portuguese accounts (early 16th century) mention **arquebus use in Deccan**, but without systematic integration into battle strategy.
5. **Absence of combined arms doctrine:** Earlier warfare lacked coordinated use of cavalry, infantry and artillery as a unified system.
   Eg: Babur's army demonstrated **combined arms approach**, unlike fragmented deployment in Sultanate armies.

Conclusion
Babur's success lay not in introducing new technology but in reimagining its use through disciplined coordination and tactical innovation. This marked the beginning of a new phase in Indian warfare where strategy began to outweigh sheer numbers.

Modern Indian history from about the middle of the eighteenth century until the present- significant events, personalities, issues.

Q. Outline the key features of Swami Vivekananda's philosophy. Evaluate their contribution to India's socio-cultural regeneration. (10 M)

Introduction
In a period marked by colonial subjugation and cultural self-doubt, **Swami Vivekananda (1863–1902)** infused Indian society with a renewed sense of spiritual confidence and social purpose. His philosophy blended **Vedantic thought with practical action**, laying the foundation for socio-cultural regeneration in modern India.

Body

<u>Key features of swami Vivekananda's philosophy</u>

1. **Practical vedanta and service to humanity:** Vivekananda emphasised that spirituality must translate into service, encapsulated in the idea of **"Daridra Narayana Seva"**.
   Eg: The establishment of **Ramakrishna Mission (1897)** institutionalised education, healthcare, and disaster relief activities, reflecting applied spirituality.
2. **Universalism and religious tolerance:** He advocated the essential unity of all religions, promoting harmony and coexistence.
   Eg: His address at the **World's Parliament of Religions, Chicago (1893)** highlighted **"acceptance of all religions as true"**, shaping global interfaith discourse.
3. **Emphasis on youth and character building:** He saw youth as the drivers of national regeneration and stressed strength, discipline, and moral integrity.
   Eg: His call to youth—**"Arise, awake and stop not till the goal is reached"**—continues to inspire programmes like **National Youth Day (12 January)** celebrated by the Government of India.
4. **Cultural nationalism rooted in spiritual identity:** Vivekananda linked India's identity to its spiritual heritage rather than political constructs.
   Eg: His speeches across India (1897–99) revived pride in **ancient Indian civilisation**, influencing leaders like **Sri Aurobindo and Subhas Chandra Bose**.
5. **Critique of social inequalities and caste rigidities:** He opposed social stagnation and emphasised upliftment of the masses.

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Eg: He stressed education for the poor and marginalised, influencing later reform movements and policies promoting **inclusive education (e.g., Article 21A, Right to Education Act 2009)**.

Contribution to socio-cultural regeneration

1. **Revival of cultural self-confidence:** His reinterpretation of Indian philosophy countered colonial narratives of inferiority.
   Eg: His global recognition post-1893 strengthened India's intellectual standing, fostering national pride.
2. **Promotion of social service as a national duty:** He transformed spirituality into a tool for nation-building.
   Eg: Organisations inspired by his ideals continue welfare work; **Ramakrishna Mission's disaster relief efforts** are widely recognised.
3. **Strengthening national integration through unity:** His message of oneness reduced sectarian divisions.
   Eg: His emphasis on unity aligns with constitutional values like **Article 51A(e)** promoting harmony and brotherhood.
4. **Influence on freedom movement ideology:** His ideas shaped the intellectual foundation of nationalism.
   Eg: Leaders like **Mahatma Gandhi** acknowledged Vivekananda's role in awakening India's spirit.
5. **Foundation for modern spiritual-humanist discourse:** He bridged tradition with modern rationality.
   Eg: Contemporary initiatives like **International Yoga Day (since 2015)** reflect the global acceptance of Indian spiritual thought rooted in similar philosophy.

Conclusion

Swami Vivekananda's philosophy transformed India's spiritual heritage into a dynamic force for social change and national awakening. His ideas continue to guide India's journey towards an inclusive, confident, and culturally rooted modern society.

Q. "Ambedkar's approach to women's rights was revolutionary in both scope and intent." Discuss. (10 M)

Introduction

The early 20th century social reform discourse in India largely sidelined women's structural rights, confining them to welfare concerns. Against this backdrop, **B. R. Ambedkar** advanced a rights-based, legal and institutional framework for women's emancipation, making his approach both transformative and enduring.

Body

Revolutionary scope of Ambedkar's approach

1. **Legal equality through constitutional guarantees:** Ambedkar ensured enforceable equality by embedding gender justice within the Constitution.
   Eg: Article 14, 15(1), 15(3) and **16** guarantee equality and allow affirmative action for women; upheld in **Air India v. Nergesh Meerza (1981)** reinforcing gender equality in service conditions.

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2. **Labour welfare and maternity protection:** As Labour Member (1942–46), he introduced progressive labour reforms benefiting women workers.
   Eg: Advocacy for maternity relief culminated in later laws like **Maternity Benefit Act, 1961**; linked to **Article 42** on humane working conditions (Source: Constitution of India).
3. **Property and inheritance rights through Hindu code bill (1948):** He attempted to radically transform patriarchal property relations.
   Eg: Though initially stalled, reforms later shaped **Hindu Succession Act, 1956**, amended in **2005** granting daughters equal coparcenary rights (Source: Ministry of Law and Justice).
4. **Access to public spaces and resources:** He linked women's dignity with broader social equality struggles.
   Eg: Mahad Satyagraha (1927) symbolically asserted equal access, inspiring later gender-inclusive interpretations of public rights (Source: NCERT Modern India).
5. **Political participation and representation:** He supported inclusive democratic participation of women.
   Eg: Universal adult franchise in **1950 Constitution** ensured equal political rights; later strengthened by **73rd and 74th Amendments (1992)** providing reservation for women in local bodies.

Revolutionary intent behind Ambedkar's vision

1. **Annihilation of caste linked with gender emancipation:** He saw caste and patriarchy as interlinked systems.
   Eg: In **Annihilation of Caste (1936)**, he argued caste endogamy sustains female oppression, influencing contemporary intersectional feminist discourse.
2. **State-led social reform over social conservatism:** He prioritised legislative intervention over gradual social change.
   Eg: Resigned as Law Minister in **1951** over dilution of Hindu Code Bill, showing commitment to structural reform (Source: Constituent Assembly Debates).
3. **Education as tool of empowerment:** He emphasised intellectual liberation of women.
   Eg: His call to **"Educate, Agitate, Organise"** led to expansion of Dalit women's education; aligns with **Article 21A** and NEP 2020 focus on gender inclusion (Source: Ministry of Education).
4. **Recognition of dignity and individual autonomy:** He framed women's rights within broader human dignity.
   Eg: Reflected in **Article 21** jurisprudence; expanded in **Justice K.S. Puttaswamy (2017)** affirming dignity and autonomy including gender rights.
5. **Advocacy for uniform civil code for gender justice:** He viewed personal laws as barriers to equality.
   Eg: Article 44 DPSP reflects this vision; contemporary debates seen in **Law Commission reports** on reforming personal laws (Source: Law Commission of India).

Conclusion

Ambedkar's vision transformed women's rights from social charity to constitutional entitlement rooted in dignity and equality. Its continued relevance lies in deepening substantive equality through law, policy and social transformation.

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Q. Examine the philosophical and political foundations of Bose's vision of independence. Analyse how it extended beyond political freedom. Evaluate its significance for post-colonial nation-building. (15 M)

Introduction

Subhas Chandra Bose's idea of independence went far beyond political sovereignty to encompass a comprehensive transformation of society. His vision integrated philosophical depth with pragmatic state-building aimed at equality, justice, and national regeneration.

Body

Philosophical and political foundations of Bose's vision of independence

1. **Synthesis of spirituality and dialectics:** Bose combined **Vedantic thought with Hegelian dialectics**, viewing reality as dynamic and shaped through conflict and synthesis, thereby justifying active political engagement.
   Eg: In **"An Indian Pilgrim" (1937)**, Bose moved away from absolute idealism and stressed a **pragmatic, action-oriented approach**, aligning philosophy with revolutionary practice.
2. **Doctrine of Samyavada (harmonious equality):** He articulated an indigenous framework blending **social justice with national unity**, avoiding blind imitation of Western ideologies.
   Eg: In **"The Indian Struggle" (1934)**, Bose proposed a synthesis of **competing ideologies like Fascism and Communism** to suit Indian socio-political realities.
3. **Primacy of a strong central state:** Bose advocated a **powerful central authority** to undertake rapid reconstruction and prevent fragmentation during transition.
   Eg: His idea of a **temporary centralised authority** during reconstruction reflected contemporary global trends while aiming at long-term mass empowerment.
4. **Commitment to social and economic equality:** Bose defined independence as ensuring **equity across class, caste, gender, and community**, not merely political freedom.
   Eg: His 1929 vision of independence included **equal distribution of wealth and removal of social injustice**, resonating with later constitutional guarantees under **Articles 14 and 15**.
5. **Scientific planning and industrialisation:** Bose emphasised **state-led industrial growth and economic planning** as essential for national strength and poverty eradication.
   Eg: As Congress President in **Haripura Session (1938)**, he supported **National Planning Committee initiatives**, anticipating planned economic development in post-independence India.

How Bose's vision extended beyond political freedom

1. **Economic emancipation as core objective:** Independence meant **ending poverty and restructuring economic relations**, not just ending colonial rule.
   Eg: Bose advocated **social ownership and control of production**, anticipating later policies like **public sector-led industrialisation in early Five-Year Plans**.
2. **Social transformation and removal of hierarchies:** He sought to dismantle **caste divisions, communalism, and gender inequalities** as part of freedom.
   Eg: His inclusive idea of independence emphasised **unity beyond caste and religion**, aligning with later reforms like **abolition of untouchability under Article 17**.
3. **Cultural and intellectual regeneration:** Bose viewed independence as revival of **national confidence and cultural synthesis** between East and West.
   Eg: His emphasis on blending **Indian spirituality with modern scientific thought** anticipated India's post-independence pursuit of **scientific temper (Article 51A(h))**.
4. **State as an instrument of mass welfare:** Freedom required a state committed to **collective welfare rather than elite interests**.

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Eg: His idea of governance as a **servant of the masses** is reflected in later welfare-oriented provisions under **Directive Principles (Part IV)**.

5. **Holistic conception of individual freedom:** Bose linked national freedom with **individual dignity, rights, and opportunities across all sections**.
   Eg: His emphasis on equality for **men and women, rich and poor** finds resonance in **constitutional guarantees of equality and justice in the Preamble**.

Significance for post-colonial nation-building

1. **Foundation for planned economic development:** Bose's emphasis on planning influenced India's adoption of **centralised economic planning mechanisms**.
   Eg: Establishment of **Planning Commission (1950)** reflected the idea of **state-led development and resource allocation**.
2. **Inspiration for welfare state model:** His vision contributed to the idea of a **state committed to social justice and redistribution**.
   Eg: Expansion of schemes like **MGNREGA (2005)** reflects continued focus on **inclusive growth and poverty alleviation**.
3. **Reinforcement of secular and inclusive nationalism:** Bose's inclusive approach strengthened the idea of **unity in diversity in nation-building**.
   Eg: Policies promoting **linguistic and cultural autonomy under the Constitution** reflect this inclusive vision.
4. **Balancing strong state with democratic evolution:** His idea of a strong state highlighted the need for **capacity in governance during early nation-building phases**.
   Eg: India's early post-independence period saw **strong central leadership under Nehru**, ensuring stability and coordinated development.
5. **Integration of ethics with statecraft:** Bose's emphasis on moral purpose ensured that political power was linked to **ethical responsibility and public welfare**.
   Eg: Contemporary governance frameworks emphasising **transparency and accountability (e.g., RTI Act, 2005)** reflect this ethical orientation.

Conclusion
Bose's vision of independence laid the foundation for a transformative and inclusive nation-state anchored in justice and development. Its enduring relevance lies in guiding India's pursuit of equitable growth and ethical governance in a changing world.

The Freedom Struggle — its various stages and important contributors/contributions from different parts of the country.

Q. Examine the key contributions of B.R. Ambedkar to labour welfare in colonial India. Analyse their significance for modern labour legislation. (10 M)

Introduction
The transformation of labour from a neglected economic factor to a rights-bearing social entity in India owes much to visionary leadership in the colonial period. Among them, **B.R. Ambedkar** played a decisive role in embedding welfare, dignity and justice into labour policy.

Body

Key contributions of B.R. Ambedkar to labour welfare

1. **Reduction of working hours:** He introduced the **8-hour workday (1942–44)** replacing the earlier 12-hour regime, marking a major shift towards humane working conditions.

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Eg: India aligned with **international labour standards (ILO norms)**; even today, the **Factories Act, 1948** continues this principle ensuring regulated working hours.

2. **Institutionalisation of social security:** He laid the foundation for schemes like **provident fund, employee insurance and welfare funds** for workers.
   Eg: The later **Employees' State Insurance Act, 1948** reflects this vision, providing **health insurance and benefits** to industrial workers (**Source: Ministry of Labour & Employment**).
3. **Women-centric labour reforms:** He strengthened **maternity benefits and wage protection for women workers**, recognising gender-specific labour vulnerabilities.
   Eg: The evolution into the **Maternity Benefit Act, 1961 (amended 2017)** ensures **paid leave and job security**, reflecting Ambedkar's early advocacy.
4. **Tripartite labour mechanism:** He institutionalised dialogue between **government, employers and workers** through the **Tripartite Labour Conference (1942)**.
   Eg: This model continues in bodies like the **Indian Labour Conference**, promoting **collective bargaining and participatory governance**.
5. **Evidence-based policymaking:** Establishment of the **Labour Investigation Committee (1944)** created a data-driven basis for reforms on wages and working conditions.
   Eg: Contemporary labour codes rely on **periodic labour surveys (PLFS, NSSO)**, reflecting the continued importance of empirical inputs (**Source: MoSPI**).
6. **Improvement in working conditions and dignity:** Measures like **housing, medical facilities and separate amenities for workers** enhanced overall welfare and dignity.
   Eg: Provisions under the **Occupational Safety, Health and Working Conditions Code, 2020** ensure **safe workplaces and welfare facilities**.

Significance for modern labour legislation

1. **Foundation for constitutional vision:** His ideas shaped **Directive Principles of State Policy**, especially **Article 39 and Article 43**, emphasising livelihood, equal pay and decent working conditions.
   Eg: Policies like **Minimum Wages Act, 1948** derive legitimacy from these constitutional directives ensuring **fair wages**.
2. **Shift towards welfare state model:** His reforms moved India away from laissez-faire to a **state-led welfare approach in labour governance**.
   Eg: The consolidation into **Four Labour Codes (2020)** reflects continued state responsibility for **social security and labour welfare**.
3. **Integration with global standards:** His policies aligned India with **ILO principles**, ensuring internationally comparable labour protections.
   Eg: India's commitment to **ILO conventions on working hours and social security** continues to shape labour reforms (**Source: ILO India reports**).
4. **Recognition of labour dignity and rights:** Labour was treated not merely as an economic input but as a **human entity with rights and dignity**.
   Eg: Recent emphasis on **gig and platform workers in Code on Social Security, 2020** reflects expansion of this rights-based approach.

Conclusion

Ambedkar's labour reforms transformed the ethical and institutional foundations of labour governance in India. His legacy endures as India continues to balance economic growth with dignity and justice for its workforce.

Post-independence consolidation and reorganization within the country.

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Q. Trace the evolution of the Naxalite movement in post-independence India. Discuss the factors responsible for its decline in recent years. (10 M)

Introduction

The Naxalite movement, emerging from agrarian discontent and socio-economic inequalities, has remained one of India's longest-running internal challenges. Its trajectory reflects the interplay between state response, development deficits, and ideological mobilization in post-independence India.

Body

Evolution of the Naxalite movement in post-independence India

1. **Origin in agrarian unrest (1967):** The movement began with the **Naxalbari uprising in West Bengal (1967)** led by **Charu Majumdar and Kanu Sanyal**, rooted in land alienation and exploitation of peasants.
   Eg: The **Naxalbari rebellion (1967)** marked the beginning of armed peasant struggle inspired by **Maoist ideology**, highlighting failure of land reforms in early post-independence India.
2. **Expansion and fragmentation (1970s–1980s):** The movement spread to states like **Andhra Pradesh, Bihar, and Jharkhand**, but fragmented into multiple groups due to ideological and leadership differences.
   Eg: Formation of groups like **People's War Group (1980)** in Andhra Pradesh and **MCC (Maoist Communist Centre)** in Bihar reflected decentralised insurgency.
3. **Consolidation under CPI (Maoist) (2004):** Merger of major factions like **PWG and MCC in 2004** led to formation of **CPI (Maoist)**, strengthening organisational and military capabilities.
   Eg: Post-2004, the movement expanded across the **"Red Corridor" spanning Chhattisgarh, Jharkhand, Odisha**, indicating peak influence.
4. **Peak of insurgency and state response (2005–2010):** Intensification of violence led to strong counter-insurgency operations and policy interventions by the state.
   Eg: Incidents like **Dantewada attack (2010)** prompted coordinated security responses and classification of LWE as **"gravest internal security threat" by PM Manmohan Singh (2006)**.

Factors responsible for decline in recent years

1. **Enhanced security operations and coordination:** Improved coordination between **Central Armed Police Forces and state police** weakened Maoist military capabilities.
   Eg: According to **Ministry of Home Affairs data**, LWE violence has declined by nearly **90% between 2009–2023**, reflecting success of integrated security strategy.
2. **Developmental outreach and infrastructure expansion:** Focus on roads, connectivity, and welfare schemes reduced isolation of affected regions.
   Eg: Schemes like **Road Requirement Plan for LWE areas** and **Aspirational Districts Programme (NITI Aayog)** improved governance penetration in Bastar and adjoining regions.
3. **Surrender and rehabilitation policy:** Incentivised surrender of cadres weakened organisational strength and leadership continuity.
   Eg: As per **MHA (2025)**, over **2,300 Maoists surrendered in 2025 alone**, indicating erosion of cadre strength.

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4. **Declining ideological appeal among tribals:** Improved state presence and awareness reduced support base among local populations.
   Eg: Expansion of **PDS, education, and mobile connectivity in tribal belts** reduced dependence on Maoist parallel governance structures.
5. **Targeted governance and legal measures:** Implementation of rights-based legislations addressed some structural grievances.
   Eg: Laws like **Forest Rights Act, 2006 and PESA Act, 1996** aimed to empower tribal communities, reducing grievances exploited by Maoists.

Conclusion
The decline of Naxalism reflects both effective state intervention and changing socio-economic dynamics. However, sustaining peace requires addressing root causes through inclusive development and just governance to prevent resurgence.

Q. "The nationalisation of banks in 1969 marked a decisive shift towards a socialist pattern of development in India". Analyse its impact on financial inclusion. (10 M)

Introduction
Post-independence India's development strategy increasingly relied on State intervention to correct structural inequalities. The nationalisation of major banks on **19 July 1969** under **Indira Gandhi** marked a decisive shift by aligning financial institutions with social and developmental goals.

Body

Shift towards socialist pattern of development

1. **State control over commanding heights:** Nationalisation brought major banks under public ownership, enabling the State to direct credit as per planned economic priorities.
   Eg: Nationalisation of **14 major banks in 1969** ensured that banking decisions were aligned with **Five-Year Plan objectives** and public welfare goals.
2. **Reorientation of credit towards social sectors:** Banking policy shifted from profit-maximisation to welfare-oriented lending for agriculture, small industries and weaker sections.
   Eg: Introduction of **priority sector lending (1972, RBI)** mandated banks to lend to **agriculture, MSMEs and vulnerable groups**, institutionalising social banking.
3. **Curtailment of concentration of wealth:** Nationalisation reduced the dominance of large industrial houses over credit allocation and financial resources.
   Eg: Pre-1969, credit was concentrated among **big business houses**, whereas post-nationalisation, lending was diversified across sectors and regions.
4. **Integration of banking with planned development:** Banks became instruments for implementing government policies and socio-economic programmes.
   Eg: Expansion of **Directed Credit Programmes** like **Lead Bank Scheme (1969)** ensured district-level planning and credit deployment.
5. **Promotion of distributive justice:** Nationalisation aimed to reduce socio-economic inequalities by ensuring equitable access to financial resources.
   Eg: Policies supporting **weaker sections lending targets** ensured that marginal farmers and small borrowers gained institutional support.

Impact on financial inclusion

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1. **Expansion of rural banking network:** Nationalisation led to rapid branch expansion in rural and previously unbanked regions.
   Eg: Bank branches increased from about **8,000 (1969)** to over **60,000 by 1990**, with a major share in **rural and semi-urban areas (RBI data)**.
2. **Enhanced access to institutional credit:** Formal credit became accessible to farmers, small entrepreneurs and self-employed groups.
   Eg: Schemes like **Integrated Rural Development Programme (IRDP, 1978)** enabled **subsidised credit to rural poor**, reducing reliance on moneylenders.
3. **Foundation for inclusive banking architecture:** Nationalisation created the institutional base for later financial inclusion initiatives.
   Eg: Pradhan Mantri Jan Dhan Yojana (2014) leveraged **public sector banks**, achieving over **50 crore bank accounts (Ministry of Finance)**.
4. **Reduction in regional disparities:** Directed lending policies improved credit flow to backward and underserved regions.
   Eg: Establishment of **Regional Rural Banks (1975)** expanded banking outreach in **remote and underdeveloped districts**.
5. **Promotion of priority sector and social banking:** Institutional mechanisms ensured continuous focus on inclusion and equity in credit delivery.
   Eg: Priority sector targets (40% of bank credit) ensured sustained lending to **agriculture, MSMEs and weaker sections (RBI guidelines)**.

Conclusion

Bank nationalisation transformed banking into a tool of inclusive development rather than mere profit-making. Its legacy continues to underpin India's evolving financial inclusion framework in a liberalised economy.

Q. "Post-independence India's linguistic policy evolved through negotiation rather than imposition." Analyse the challenges faced during the transition to official language policy. Discuss the role of political movements in shaping outcomes. (15 M)

Introduction

India's linguistic diversity posed one of the most complex challenges to nation-building after 1947. The evolution of language policy reflects a careful balancing of **unity, federal accommodation and democratic negotiation** rather than rigid imposition.

Body

<u>Linguistic policy evolved through negotiation rather than imposition</u>

1. **Constitutional compromise on official language:** The Constitution adopted **Hindi in Devanagari script as official language (Article 343, 1950)** while allowing continued use of English for 15 years, reflecting a negotiated settlement.
   Eg: The **Munshi-Ayyangar formula (1949)** ensured coexistence of **Hindi and English**, avoiding immediate imposition and accommodating non-Hindi states
2. **Extension of English through legislative accommodation:** The **Official Languages Act, 1963** and its amendment in **1967** ensured continued use of English alongside Hindi beyond 1965.

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Eg: Following protests, Parliament guaranteed that **English would continue indefinitely for official purposes**, reflecting political accommodation

3. **Federal flexibility in language use:** The Constitution under **Article 345** allowed states to adopt their own official languages, ensuring decentralised linguistic autonomy.
   Eg: States like **Tamil Nadu continued using Tamil and English**, resisting pressure to adopt Hindi, demonstrating negotiated federalism

Challenges during transition to official language policy

1. **Diversity and regional linguistic identities:** India's vast linguistic plurality made the choice of a single official language politically sensitive.
   Eg: The presence of **22 scheduled languages (Eighth Schedule)** reflects diversity, making uniform linguistic policy difficult
2. **Fear of cultural domination by Hindi:** Non-Hindi speaking regions perceived Hindi imposition as cultural and political marginalisation.
   Eg: In **Tamil Nadu**, fears of **"Hindi imperialism"** led to widespread resistance, especially during the 1960s
3. **Administrative and functional constraints:** Immediate replacement of English posed challenges in governance, judiciary and higher education.
   Eg: Supreme Court proceedings continue in English (Article 348), indicating practical necessity of retaining English
4. **Political resistance and unrest:** Language transition triggered mass protests, sometimes violent, complicating policy implementation.
   Eg: The **1965 anti-Hindi agitations** saw widespread protests and forced the Centre to reconsider its stance
5. **Balancing national integration with federalism:** The Centre had to ensure unity without alienating states, making policy formulation complex.
   Eg: The **States Reorganisation Act, 1956** based on linguistic lines further reinforced regional identities, complicating national language debates

Role of political movements in shaping outcomes

1. **Dravidian movement and anti-Hindi agitations:** Regional movements actively resisted Hindi imposition and shaped policy reversals.
   Eg: The **DMK-led protests in Tamil Nadu (1965)** compelled the Centre to assure continued use of English and revise its approach
2. **Mass mobilisation influencing policy decisions:** Popular protests forced democratic accountability and policy reconsideration.
   Eg: Following agitations, **Prime Minister Lal Bahadur Shastri (1965)** assured continuation of English, reflecting responsiveness to public pressure
3. **Regional parties strengthening federal bargaining:** Political rise of regional parties ensured stronger representation of linguistic interests.
   Eg: The **DMK's electoral victory in 1967** marked the decline of Congress dominance in Tamil Nadu and strengthened federal negotiations
4. **Institutionalisation of linguistic pluralism:** Movements contributed to formal recognition of multiple languages in governance.

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Eg: Expansion of the **Eighth Schedule over time (now 22 languages)** reflects accommodation of linguistic aspirations

5. **Shaping cooperative federalism in language policy:** Political negotiations transformed language policy into a consensus-driven framework.
   Eg: Continued use of **English for inter-state and Centre-state communication** reflects a negotiated and pragmatic federal arrangement

Conclusion

India's language policy demonstrates that **democratic negotiation is central to managing diversity in a plural society**. Sustaining this balance requires continued sensitivity to regional identities while strengthening cooperative federalism.

History of the world will include events from 18th century such as industrial revolution, world wars, redrawal of national boundaries, colonization, decolonization, political philosophies like communism, capitalism, socialism etc.— their forms and effect on the society.

Q. What were the major causes behind the 1973 oil crisis? Assess its impact on the industrial economies of the West. (10 M)

Introduction

The 1973 oil crisis marked a turning point in global economic history, exposing the vulnerability of industrial economies to energy shocks. It demonstrated how geopolitical tensions could directly disrupt economic systems.

Body

<u>Causes of the 1973 oil crisis</u>

1. **OPEC oil embargo during Yom Kippur War (1973):** Arab members of OPEC imposed an oil embargo against nations supporting Israel, especially the United States, causing an abrupt supply disruption.
   Eg: In **October 1973**, countries like **Saudi Arabia and Kuwait cut production and restricted exports to the US**, triggering a global supply shock.
2. **Assertion of resource control by oil-producing nations:** OPEC countries sought to assert sovereignty over natural resources and gain pricing power after prolonged Western dominance.
   Eg: The coordinated action by **OPEC (established 1960)** enabled collective bargaining and production cuts to influence global markets.
3. **Breakdown of Bretton Woods system (1971):** The US decision to end dollar convertibility to gold destabilised the monetary system, affecting oil revenues.
   Eg: The **Nixon Shock (1971)** led to dollar depreciation, reducing earnings of oil exporters and prompting price hikes.
4. **Rising demand and import dependence in the West:** Rapid industrial growth increased dependence on imported oil without adequate domestic capacity.
   Eg: The **US oil production peaked in 1970**, increasing reliance on imports from West Asia.

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Impact on industrial economies of the West

1. **Stagflation and economic crisis:** Simultaneous inflation and stagnation disrupted conventional economic management frameworks.
   Eg: In **1974–75, the US and UK experienced high inflation with rising unemployment**, marking a major stagflation phase.
2. **Energy policy shifts and conservation measures:** Western nations adopted long-term strategies to reduce dependence on oil imports.
   Eg: The US introduced **fuel rationing, speed limits and created the Strategic Petroleum Reserve in 1975**.
3. **Decline in industrial productivity and growth:** Increased energy costs raised production costs and reduced competitiveness.
   Eg: Western European industries faced higher input costs, leading to slowdown in manufacturing output.
4. **Strengthening of dollar-based oil trade system:** Oil pricing in dollars increased global dependence on US currency.
   Eg: The **US–Saudi agreement (1974–75)** institutionalised oil trade in dollars, reinforcing US financial influence.

Conclusion

The crisis highlighted how energy security became central to economic stability in industrial economies. It reshaped global economic structures and reinforced the strategic role of resources in world history.

Q. Examine the significance of the United States' entry into World War I. Analyse its impact on the outcome of the war. (10 M)

Introduction

The entry of the **United States on April 6, 1917** transformed World War I from a prolonged stalemate into a decisive conflict. It marked the shift of global power and injected fresh momentum into the Allied war effort.

Body

Significance of the United States' entry into World War I

1. **Breakdown of prolonged stalemate:** The Western Front had remained deadlocked since 1914, but U.S. entry altered the strategic equilibrium by strengthening Allied capacity.
   Eg: By **mid-1918, over 10 lakh American troops (American Expeditionary Forces)** had reached Europe, boosting Allied offensives such as the **Second Battle of the Marne (1918)**.
2. **Psychological and moral boost to Allies:** U.S. participation revived Allied morale at a time when war fatigue and losses had weakened their resolve.
   Eg: The entry followed the **Zimmermann Telegram (1917)** and **unrestricted submarine warfare**, galvanising public and military confidence in **Britain and France**.
3. **Economic and industrial superiority:** The U.S. provided critical financial and material support, ensuring sustained Allied war capability.

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Eg: Massive **war loans and supplies from the U.S. economy**, which had already been supporting Allies since **1915**, ensured uninterrupted logistics.

4. **Shift in global diplomatic balance:** U.S. entry internationalised the conflict further and signalled the decline of European dominance in global affairs.
   Eg: President Woodrow Wilson's leadership positioned the U.S. as a key diplomatic actor in shaping post-war negotiations.

Impact on the outcome of the war

1. **Decisive military advantage to Allies:** Fresh troops and resources enabled the Allies to launch coordinated offensives that Germany could not sustain against.
   Eg: The **Hundred Days Offensive (August–November 1918)** saw Allied forces, supported by U.S. troops, push back German forces leading to collapse.
2. **Acceleration of German defeat:** Facing internal unrest and inability to match Allied strength, Germany sought an armistice.
   Eg: The **Armistice of November 11, 1918** followed German military exhaustion and political instability after failed offensives in 1918.
3. **Influence on post-war settlement:** The U.S. shaped the ideological framework of peace, emphasising liberal internationalism.
   Eg: Wilson's Fourteen Points (January 1918) influenced the **Treaty of Versailles (1919)** and the creation of the **League of Nations**.
4. **Emergence of the United States as a global power:** The war marked the transition of the U.S. from relative isolation to a central role in global politics.
   Eg: The U.S. emerged as a major **creditor nation post-1918**, replacing Europe's pre-war financial dominance.

Conclusion

The U.S. entry not only hastened Allied victory but also reconfigured global power structures. It laid the foundation for a new international order where the United States emerged as a decisive actor in world affairs.

Q. Identify the key socio-economic conditions in China prior to 1949. Bring out how these conditions contributed to large-scale revolutionary mobilisation. (10 M)

Introduction

Early 20th century China witnessed **deep agrarian distress, political fragmentation and foreign domination**, creating a volatile social base.
These structural conditions generated widespread discontent, which became fertile ground for **mass revolutionary mobilisation under the Chinese Communist Party**.

Body

Socio-economic conditions in China before 1949

1. **Semi-feudal agrarian structure and land inequality:** A vast majority of peasants were landless or marginal, facing high rents and exploitation by landlords.

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Eg: In pre-1949 China, a small landlord class controlled large tracts of land while peasants paid **exorbitant rents and taxes**, leading to chronic poverty and unrest.

2. **Warlordism and political fragmentation after 1911:** The fall of the Qing dynasty in **1911** led to weak central authority and rise of regional warlords exploiting local populations.
   Eg: During the **Warlord Era (1916–1928)**, peasants suffered forced taxation and conscription, deepening resentment against existing order.
3. **Impact of foreign imperialism and unequal treaties:** Western powers and Japan exercised economic and territorial control, undermining China's sovereignty.
   Eg: The **Treaty of Versailles (1919)** transferring German rights in Shandong to Japan triggered the **May Fourth Movement (1919)**, reflecting nationalist anger.
4. **Industrial backwardness and urban worker exploitation:** Limited industrialisation led to poor working conditions and low wages for a small but significant working class.
   Eg: Workers in treaty port cities like **Shanghai faced long working hours and low wages**, fostering support for socialist ideas.
5. **Economic crisis and hyperinflation during civil war:** Prolonged conflict between KMT and CCP weakened the economy and eroded living standards.
   Eg: By the **late 1940s**, inflation under the **Kuomintang government** severely reduced purchasing power, alienating urban middle classes.

How these conditions facilitated revolutionary mobilisation

1. **Peasant discontent as revolutionary base:** Agrarian exploitation enabled the CCP to mobilise peasants through promises of land redistribution.
   Eg: The CCP's **land reform programmes in Jiangxi Soviet (1930s)** redistributed land, gaining massive peasant support.
2. **Nationalist sentiment against foreign domination:** Anti-imperialist feelings unified diverse groups behind revolutionary forces.
   Eg: The **May Fourth Movement (1919)** mobilised students and intellectuals, later aligning with Communist ideology.
3. **Delegitimisation of KMT rule:** Corruption, inefficiency and inability to address economic distress weakened KMT's legitimacy.
   Eg: Failure of KMT to control **inflation and warlord excesses in 1940s** led to loss of urban and rural support.
4. **Mobilisation through guerrilla and mass-based strategies:** CCP adopted rural-based strategies suited to Chinese conditions.
   Eg: The **Long March (1934–35)** strengthened CCP organisation and spread revolutionary ideology among rural masses.

Conclusion

The Chinese Revolution was rooted not merely in ideology but in **deep socio-economic contradictions of Chinese society**. By aligning revolutionary goals with popular grievances, the CCP transformed discontent into a **mass political movement that reshaped China's destiny**.

Salient features of Indian Society, Diversity of India.

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Q. "Caste endogamy remains the bedrock of social organisation in India despite modernisation." Critically examine. (15 M)

India's social structure continues to exhibit a paradox where **modernisation coexists with deep-rooted traditional institutions**, with marriage emerging as a key site where this tension is most visible. Despite economic mobility and urbanisation, **caste endogamy continues to reproduce social hierarchies across generations**.

Body

Caste endogamy as bedrock of social organisation

1. **Marriage as mechanism of caste reproduction: Caste endogamy ensures continuity of caste boundaries across generations by restricting social mixing**. It preserves ritual status and hereditary occupation linkages within caste groups.
   Eg: According to **India Human Development Survey (IHDS-II, 2011-12)**, over **90–95% marriages in India are within the same caste**, indicating persistence of endogamy (**Source: IHDS**).
2. **Kinship networks and social capital: Endogamous marriages reinforce intra-caste kinship ties, enabling access to social support, employment and political patronage**. This strengthens caste-based solidarity.
   Eg: Dominant castes like **Jats, Patidars and Marathas** use marriage alliances to consolidate **economic and political networks**, visible in local electoral mobilisations (**Source: Sociological studies on caste politics**).
3. **Control over property and inheritance: Endogamy ensures that land, wealth and resources remain within caste groups, maintaining economic hierarchies**. It prevents fragmentation of caste-based assets.
   Eg: In rural India, **land-owning castes** prefer intra-caste marriages to retain **agricultural land within lineage groups**, as observed in **NSSO landholding patterns**.
4. **Patriarchy and control over women: Caste endogamy is closely linked to patriarchal control over women's sexuality and marital choices**. Women's choices are regulated to preserve caste purity.
   Eg: Cases of **honour killings reported by NCRB (Crime in India Reports)** reflect violence against women marrying outside caste, showing enforcement of endogamy.
5. **Social sanctions and informal institutions: Community bodies like khap panchayats enforce caste endogamy through social pressure, ostracism or violence**. This informal regulation sustains the system.
   Eg: The **Supreme Court in Shakti Vahini vs Union of India (2018)** condemned khap interference in marriages and issued preventive guidelines against honour crimes.

Limits of modernisation in weakening caste endogamy

1. **Urbanisation without social transformation: Migration to cities has not fully dismantled caste identities, as matrimonial choices still rely on caste networks**. Modern spaces replicate traditional preferences.

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Eg: Popular matrimonial platforms and advertisements explicitly filter matches by **caste categories**, indicating persistence even among urban educated groups.

2. **Education and economic mobility reinforcing caste: Instead of eroding caste, education often leads to 'caste consolidation' where similar-status groups prefer intra-caste alliances**.
   Eg: Studies show **middle-class upwardly mobile groups** prefer endogamy to maintain **status parity**, reflecting what **M.N. Srinivas described as 'Sanskritisation' dynamics**.

Factors indicating gradual weakening

1. **Rise of inter-caste marriages: Though limited, inter-caste unions are increasing due to urban exposure, education and individual agency**. This reflects cracks in rigid endogamy.
   Eg: Government schemes like **Dr. Ambedkar Scheme for Social Integration through Inter-Caste Marriage (2013)** incentivise such marriages (**Source: Ministry of Social Justice and Empowerment**).
2. **Constitutional morality and judicial protection: Legal framework increasingly prioritises individual choice over social norms, challenging caste endogamy**.
   Eg: In **Shafin Jahan vs Asokan K.M. (2018)**, the **Supreme Court upheld the right to choose a spouse under Article 21**, reinforcing autonomy.
3. **Changing aspirations and youth agency: Younger generations are increasingly asserting personal choice in marriage, especially in urban and semi-urban areas**.
   Eg: Surveys like **Lokniti-CSDS youth studies** indicate rising acceptance of **love marriages**, though still constrained by caste considerations.

Way forward

1. **Strengthening constitutional morality: Promote values of liberty, equality and dignity through education and public discourse to counter caste-based social norms**.
   Eg: The **Supreme Court in Navtej Singh Johar (2018)** emphasised **constitutional morality over social morality**, which can guide societal change.
2. **Incentivising inter-caste marriages: Expand financial and social support schemes to encourage social integration and reduce caste barriers**.
   Eg: Enhancing coverage and awareness of **Dr. Ambedkar Inter-Caste Marriage Scheme** can improve uptake (**Source: Ministry of Social Justice and Empowerment**).
3. **Community-level sensitisation: Engage local institutions, civil society and youth groups to challenge regressive practices and promote social acceptance**.
   Eg: Awareness campaigns by **State Social Welfare Departments** in states like **Tamil Nadu** have improved reporting and acceptance.
4. **Strict enforcement against honour crimes: Ensure effective implementation of Supreme Court guidelines and fast-track prosecution to deter violence**.
   Eg: Directions in **Shakti Vahini case (2018)** mandate preventive, remedial and punitive measures against honour crimes.

Conclusion

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While caste endogamy continues to structure social relations, its hold is **gradually eroding under the influence of constitutional values and social change**. A sustained push combining **legal enforcement, social reform and generational shifts** can accelerate this transformation.

Q. "Family bonds in India are sustained as much by cultural expectations as by emotional realities." Examine this statement. Analyse the factors leading to increasing familial estrangement. (10 M)

Introduction

Indian family structures have historically been anchored in deeply embedded cultural norms that prioritise duty, hierarchy and continuity. However, the growing assertion of individual autonomy is exposing the fragile balance between social expectations and lived emotional experiences.

Body

Family bonds sustained by cultural expectations and emotional realities

1. **Norm of filial duty and social sanction:** Indian society emphasises **filial piety** and moral obligation towards parents, reinforced through customs, religion and community pressure, often irrespective of emotional quality of relationships.
   Eg: Maintenance and Welfare of Parents and Senior Citizens Act, 2007 mandates children to provide maintenance, reflecting legal reinforcement of cultural duty; highlighted in **Ministry of Social Justice reports**.
2. **Joint family legacy and interdependence:** Traditional **joint family system** ensured economic and emotional interdependence, where cohesion was maintained more by obligation than individual choice.
   Eg: Census 2011 data indicates gradual decline in joint households, yet persistence in rural India shows continued reliance on **collective family structures**.
3. **Parental authority shaping identity:** Strong parental influence in early life shapes values and identity, making family bonds psychologically embedded even when emotionally strained.
   Eg: Studies by **NCERT on adolescence (2019)** highlight continued **parental influence in decision-making**, even in urban contexts.
4. **Marriage and kinship as social institutions:** Family bonds are sustained through **arranged marriages and kinship networks**, reinforcing continuity of relationships beyond individual preference.
   Eg: National Family Health Survey (NFHS-5, 2019–21) shows high prevalence of **family-mediated marriages**, indicating cultural continuity.

Factors leading to increasing familial estrangement

1. **Rise of individualism and autonomy:** Increasing emphasis on **self-fulfilment and personal boundaries** is redefining relationships, reducing tolerance for hierarchical or coercive family structures.
   Eg: Growing discourse around **"toxic relationships" on digital platforms** like Reddit and Instagram reflects assertion of **individual emotional well-being**.

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2. **Urbanisation and migration:** Physical separation due to jobs and education weakens daily interaction and emotional bonding within families.
   Eg: Periodic Labour Force Survey (PLFS 2022-23) shows rising **inter-state migration**, contributing to nuclearisation and distance from parents.
3. **Changing gender roles and expectations:** Women's education and employment are challenging traditional authority structures, leading to conflicts within families.
   Eg: Increased **female labour force participation (PLFS 2022-23)** is associated with renegotiation of **household power dynamics**.
4. **Digital mediation of relationships:** Social media amplifies alternative support systems, reducing dependence on family while also reshaping perceptions of relationships.
   Eg: Rise of **online communities discussing estrangement and mental health**, as noted in **UNICEF digital well-being reports**, indicates shifting social anchors.
5. **Legal and institutional shifts in family norms:** Judicial recognition of individual rights sometimes redefines traditional obligations, creating tension between law and culture.
   Eg: Allahabad High Court (2024) held that **moral responsibility cannot be enforced as legal duty**, reflecting limits of legal intervention in family matters.

Conclusion
The Indian family today stands at the intersection of tradition and transformation, where cultural expectations alone can no longer sustain relationships. A balanced approach that respects both emotional authenticity and social responsibility is essential for sustaining meaningful familial bonds.

Role of women and women's organization, population and associated issues, poverty and developmental issues, urbanization, their problems and their remedies.

Q. The persistence of customary land systems in multi-ethnic regions can both preserve identity and generate conflict. Analyse this statement. Evaluate its impact on inter-community relations. (15 M)

Introduction
In culturally diverse regions, land is not merely an economic asset but a marker of identity, belonging and historical memory. The coexistence of customary land systems with modern governance frameworks often produces both cohesion within communities and contestation between them.

Body

<u>Customary land systems as preservers of identity</u>

1. **Cultural continuity and identity preservation:** Customary systems embed land within traditions, rituals and lineage, strengthening collective identity.
   Eg: In **Nagaland**, land is largely owned by tribes and clans under customary law, protected under **Article 371A**, ensuring preservation of **Naga cultural identity** and traditional institutions.
2. **Community cohesion and social solidarity:** Collective ownership promotes intra-community cooperation and reduces internal inequality.

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Eg: Among **Khasi tribes in Meghalaya**, land is managed through clan-based systems, reinforcing **matrilineal social organisation** and strong kinship bonds.

3. **Protection of indigenous rights and autonomy:** Customary tenure safeguards tribal populations from external exploitation and displacement.
   Eg: The **Sixth Schedule (Articles 244(2) and 275(1))** provides Autonomous District Councils in Northeast India to protect **tribal land rights and governance autonomy**.
4. **Sustainable resource management practices:** Indigenous systems often follow ecological norms suited to local conditions.
   Eg: Shifting cultivation (jhum) practices in Northeast India reflect traditional ecological knowledge, maintaining **biodiversity balance** in fragile hill ecosystems.
5. **Resistance to homogenisation under modern laws:** Customary systems act as a counterbalance to uniform legal imposition.
   Eg: The **PESA Act, 1996** recognises **Gram Sabha authority over land and resources** in Scheduled Areas, reinforcing local traditions against external interventions.

Customary land systems as sources of conflict

1. **Ambiguity in land ownership and boundaries:** Lack of codification leads to overlapping claims and disputes.
   Eg: In **Manipur hill districts**, absence of formal land records has led to **conflicts between Kuki and Naga groups over territorial claims** (Indian Express, 2026).
2. **Inter-community competition over resources:** Customary exclusivity can intensify ethnic tensions in multi-ethnic settings.
   Eg: The ongoing **Meitei–Kuki conflict (since 2023)** is partly linked to **land access and territorial control in hill areas**.
3. **Tensions between mobile and sedentary populations:** Traditional mobility patterns often clash with settled agricultural communities.
   Eg: Expansion of **Kuki settlements through village proliferation** has caused friction with **Naga groups over land usage patterns**.
4. **Incompatibility with modern development frameworks:** Infrastructure projects face resistance due to differing land regimes.
   Eg: Disputes over **compensation for highway expansion in Manipur** highlight conflict between **state acquisition norms and customary ownership claims**.
5. **Elite capture and internal inequities:** Customary systems may be dominated by traditional elites, marginalising weaker sections.
   Eg: In some tribal regions, **village chiefs control land allocation**, limiting access for **women and lower-status groups** (Xaxa Committee, 2014).

Impact on inter-community relations

1. **Reinforcement of ethnic boundaries:** Strong land-identity linkage sharpens ‘insider vs outsider’ distinctions.
   Eg: In Northeast India, land rights are often restricted to **indigenous groups**, deepening **ethnic segmentation in social relations**.
2. **Escalation of ethnic conflicts:** Competing territorial claims transform social differences into violent confrontations.

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Eg: The **Manipur ethnic conflict (2023–26)** shows how land disputes can escalate into **large-scale inter-community violence**.

3. **Erosion of trust and social cohesion:** Persistent disputes reduce scope for cooperation between communities.
   Eg: Recurrent tensions between **tribal groups in hill districts** have weakened **inter-tribal alliances and shared governance mechanisms**.
4. **Fragmentation of shared public spaces:** Segregation increases as communities retreat into exclusive territorial zones.
   Eg: In conflict-prone areas of Manipur, **villages have become ethnically homogenous**, reducing everyday social interaction.
5. **Challenges to national integration:** Localised identity conflicts complicate broader integration efforts.
   Eg: The **Second Administrative Reforms Commission (2007)** highlighted that unresolved ethnic and land issues in Northeast India hinder **inclusive nation-building**.

Conclusion

Customary land systems embody identity and continuity but can also sharpen divisions in plural societies. A balanced approach that respects tradition while ensuring equity and clarity is essential for sustainable social harmony.

Q. Examine the structural vulnerabilities of migrant workers in Indian cities. Analyse how recent crises have exposed these weaknesses. Suggest institutional reforms to address them. (15 M)

Introduction

India's urban growth is powered by invisible labour that remains structurally excluded from formal protections. Recurrent crises—from **COVID-19 (2020)** to recent **energy supply disruptions (2026)**—have exposed deep fault lines in migrant livelihoods and welfare access.

Body

<u>Structural vulnerabilities of migrant workers in Indian cities</u>

1. **Informality of employment and lack of legal protection:** A large proportion of migrants work in informal sectors without written contracts, social security or job security.
   Eg: Periodic Labour Force Survey (PLFS) shows over **80% workforce in informal sector**, leaving migrants vulnerable to sudden job loss during crises like **COVID-19 lockdown (2020)**.
2. **Exclusion from social protection due to portability gaps:** Welfare benefits are often linked to domicile, limiting access in destination cities.
   Eg: Despite **One Nation One Ration Card (ONORC, 2019)**, initial implementation gaps during **COVID-19** left migrants without food security (Source: **Ministry of Consumer Affairs reports**).
3. **Inadequate housing and living conditions:** Migrants reside in overcrowded informal settlements with poor basic services.
   Eg: The **Economic Survey 2017-18** highlighted lack of affordable rental housing; crisis situations forced reverse migration due to inability to sustain urban living.

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4. **Documentation and identity barriers:** Migrants often lack local proof of residence, restricting access to schemes like LPG, health and banking services.
   Eg: Limited awareness and documentation hindered access to **Pradhan Mantri Ujjwala Yojana (PMUY 2.0, 2021)** among migrants.
5. **Weak bargaining power and labour exploitation:** Absence of unions and regulatory oversight leads to wage suppression and unsafe working conditions.
   Eg: Inter-State Migrant Workmen Act, 1979 remains weakly enforced, leading to exploitation in sectors like construction and textiles.
6. **Gendered vulnerabilities and invisibilisation:** Women migrants face dual burden of unpaid care work and unsafe working conditions.
   Eg: Women workers in informal sectors were disproportionately affected during **COVID-19**, as noted in **ILO India reports (2021)**.
7. **Limited access to healthcare and nutrition:** Urban health systems often exclude migrants due to cost and accessibility barriers.
   Eg: National Family Health Survey-5 (2019-21) indicates poorer nutrition outcomes among migrant households.

How recent crises have exposed these weaknesses

1. **Mass reverse migration and livelihood collapse:** Sudden economic shocks triggered large-scale return migration, revealing fragile urban integration.
   Eg: COVID-19 lockdown (2020) led to one of the largest internal migrations since independence (Source: **Stranded Workers Action Network reports**).
2. **Breakdown of food and energy security:** Supply disruptions directly impacted daily survival due to lack of buffers.
   Eg: 2026 LPG shortage crisis forced migrant families in **Surat** to reduce meals and return to villages, highlighting energy-food nexus.
3. **Collapse of informal support systems:** Closure of dhabas, hostels and informal markets intensified vulnerabilities.
   Eg: During **fuel shortages**, closure of roadside eateries deprived workers of affordable food options.
4. **Rising cost of living and indebtedness:** Inflation disproportionately affected low-income migrants with no savings.
   Eg: Reports show increase in monthly expenses of migrant households due to rising food and fuel prices during recent crises.
5. **Administrative invisibility and data gaps:** Lack of real-time migrant data hindered targeted policy response.
   Eg: Absence of a comprehensive migrant database delayed relief measures during **COVID-19**, acknowledged by **NITI Aayog discussions (2020)**.

Institutional reforms to address migrant vulnerabilities

1. **Universalisation and portability of welfare schemes:** Ensure seamless access to food, health and fuel across states.
   Eg: Strengthening **ONORC** and integrating it with **Ayushman Bharat (PM-JAY, 2018)** for portability of benefits.

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2. **Implementation of labour codes with focus on migrants:** Effective enforcement of **Code on Social Security, 2020** to extend benefits to informal workers.
   Eg: Registration of gig and migrant workers on **e-Shram portal (launched 2021)** to enable targeted welfare delivery.
3. **Creation of a national migrant worker database:** Real-time data for better policy targeting and crisis response.
   Eg: Supreme Court in In Re: Problems and Miseries of Migrant Labourers (2021) directed governments to maintain migrant data.
4. **Affordable rental housing and urban inclusion policies:** Promote dignified living conditions through policy interventions.
   Eg: Affordable Rental Housing Complexes (ARHCs) scheme, 2020 under **PMAY-U** for migrant housing.
5. **Strengthening inter-state coordination mechanisms:** Institutional frameworks for portability of rights and grievance redressal.
   Eg: NITI Aayog policy papers recommend inter-state coordination cells for migrant welfare.
6. **Enhancing awareness and last-mile delivery of schemes:** Focus on outreach through NGOs and local bodies.
   Eg: Collaboration with organisations like **Aajeevika Bureau** has improved migrant access to welfare services.
7. **Legal and constitutional safeguards:** Uphold rights under **Article 14 (Equality)**, **Article 19(1)(d) (Freedom of movement)** and **Article 21 (Right to life)**.
   Eg: Supreme Court (2021 migrant labour case) emphasised state responsibility to ensure food and transport security.

Conclusion

Migrant workers are central to India's urban economy, yet remain structurally excluded from its institutional framework. Building a portable, inclusive and rights-based welfare architecture is essential to transform vulnerability into resilience.

Q. Bring out the socio-economic consequences of heat waves in India. Assess their impact on productivity and urban life. Suggest policy measures required for mitigation. (15 M)

Introduction

Heat waves in India have evolved from seasonal discomfort to a structural development challenge, affecting livelihoods, cities and human well-being. Their growing frequency reflects the intersection of **global warming and local urban vulnerabilities**, demanding urgent policy attention.

Body

<u>Socio-economic consequences of heat waves</u>

1. **Decline in labour productivity:** Extreme heat reduces work efficiency, especially in outdoor sectors like construction and agriculture, impacting economic output.
   Eg: As per **International Labour Organization (ILO, 2019)**, India is projected to lose **5.8% of working hours by 2030** due to heat stress, significantly affecting labour-intensive sectors.
2. **Public health burden escalation:** Heat waves increase cases of heatstroke, dehydration and cardiovascular stress, straining healthcare systems.

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Eg: During the **2022 India heatwave**, hospitals reported a surge in **heat-related illnesses**, particularly among vulnerable populations (Source: **Ministry of Health & Family Welfare**).

3. **Rising inequality and vulnerability:** Poor populations lack access to cooling infrastructure, making them disproportionately exposed to heat stress.
   Eg: Urban slum dwellers in cities like **Delhi and Ahmedabad** experience higher exposure due to **lack of green cover and ventilation** (Source: **National Disaster Management Authority – NDMA**).
4. **Economic stress on households:** Increased demand for cooling leads to higher electricity consumption, raising household expenditure.
   Eg: Peak electricity demand crossed **240 GW in 2024**, driven largely by cooling needs (Source: **Central Electricity Authority**).
5. **Stress on urban infrastructure:** Heat waves strain power grids, water supply systems and transport networks, reducing urban efficiency.
   Eg: Cities like **Delhi and Mumbai** reported **power outages and water shortages during peak summers** due to excessive demand.

Impact on productivity and urban life

1. **Disruption of urban work cycles:** High daytime and night-time temperatures reduce effective working hours and increase fatigue.
   Eg: Rising **minimum temperatures in Indian cities** reduce night-time cooling, affecting next-day productivity (Source: **India Meteorological Department – IMD**).
2. **Urban heat island intensification:** Built-up areas with concrete and glass amplify heat, worsening living conditions in cities.
   Eg: Bengaluru and Gurugram show increased heat retention due to **loss of water bodies and green cover** (Source: **Urban climate studies, IMD**).
3. **Social tensions and behavioural impacts:** Heat stress increases irritability, leading to higher instances of urban conflicts and road rage.
   Eg: Reports indicate **rise in road rage incidents during peak summer months** in major metros (Source: **Urban behavioural studies**).
4. **Reduced liveability of cities:** Outdoor activities decline, affecting social life, tourism and urban vibrancy.
   Eg: Decline in **public space usage and outdoor events** during heat waves in cities like **Jaipur and Delhi**.

Policy measures required for mitigation

1. **Implementation of heat action plans:** Early warning systems and coordinated response strategies reduce mortality and risk.
   Eg: Ahmedabad Heat Action Plan (2013) is a global best practice, reducing heat-related deaths (Recognised by **NDMA and WHO**).
2. **Urban greening and blue infrastructure:** Increasing green cover and restoring water bodies help regulate urban temperatures.
   Eg: Bengaluru Airport Terminal 2 greening model demonstrates climate-sensitive infrastructure design.
3. **Climate-sensitive urban planning:** Building materials, design and zoning must align with local climatic conditions.
   Eg: National Mission on Sustainable Habitat (NAPCC) promotes climate-resilient urban planning (Source: **Government of India**).
4. **Energy-efficient cooling solutions:** Promoting sustainable cooling reduces energy burden and emissions.
   Eg: India Cooling Action Plan (2019) aims to reduce cooling demand by **20-25% by 2037-38**.

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5. **Protection of vulnerable populations:** Targeted interventions like shelters, water access and work-hour regulations are essential.
   Eg: NDMA guidelines recommend **rescheduling work hours and providing cooling shelters** for outdoor workers.

Conclusion
Heat waves are redefining India's development trajectory by linking climate stress with socio-economic outcomes. A shift towards **climate-resilient urban systems and inclusive adaptation strategies** is essential to safeguard both livelihoods and liveability.

Q. "Urban planning in India often fails to internalise principles of social equity and climate resilience." Examine this statement. Analyse its implications for vulnerable communities. Suggest a framework for inclusive and climate-resilient urban development. (15 M)

Introduction
Indian cities are at the frontline of the climate crisis, yet their growth reflects deep social asymmetries. The disconnect between planning processes and lived realities of vulnerable groups has intensified risks rather than mitigating them.

Body

Failure to internalise social equity and climate resilience

1. **Spatial segregation and exclusionary zoning:** Urban planning often pushes marginalised groups to ecologically fragile zones like floodplains and marshlands.
   Eg: In **Chennai resettlement colonies like Perumbakkam**, communities relocated from central areas now face **higher flood vulnerability due to low-lying marshland locations** (Source: Chennai Climate Action Plan, 2022).
2. **Inadequate integration of climate risks in planning frameworks:** Master plans rarely incorporate localised climate risk assessments.
   Eg: The **Delhi Master Plan 2021** faced criticism for insufficient focus on **urban flooding and heat island mitigation**, despite rising extreme weather events (Source: Centre for Science and Environment reports).
3. **Weak implementation of constitutional decentralisation:** Despite **74th Constitutional Amendment Act, 1992**, Urban Local Bodies lack financial and functional autonomy.
   Eg: The **Second Administrative Reforms Commission (2007)** noted that ULBs remain dependent on state governments, limiting **context-specific climate adaptation planning**.
4. **Neglect of informal settlements in urban governance:** Slums are often treated as illegal rather than integral to urban systems.
   Eg: According to **Census 2011**, a significant share of urban poor reside in informal settlements with **limited access to drainage, sanitation and resilient infrastructure**.

Implications for vulnerable communities

1. **Heightened exposure to climate hazards:** Marginalised groups face disproportionate impacts from floods, heatwaves and cyclones.
   Eg: During **Cyclone Michaung (2023)**, low-income settlements in Chennai experienced **prolonged flooding and infrastructure collapse** due to poor planning.

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2. **Erosion of livelihood security:** Displacement and poor connectivity reduce access to employment opportunities.
   Eg: Relocated communities in **urban peripheries** often face **loss of informal sector jobs due to distance from city centres** (Source: World Bank urbanisation reports).
3. **Reduced access to basic services and relief:** Structural inequalities limit access to healthcare, water, and disaster relief.
   Eg: Studies by **National Institute of Urban Affairs (NIUA)** highlight that informal settlements receive **delayed or inadequate disaster response services**.
4. **Breakdown of social capital and coping mechanisms:** Forced resettlement disrupts community networks critical during crises.
   Eg: Peripheral resettlement in cities like **Delhi and Mumbai** has weakened **community-based support systems during extreme weather events**.

Framework for inclusive and climate-resilient urban development

1. **Integrating equity into climate planning:** Urban policies must prioritise vulnerable groups in climate adaptation strategies.
   Eg: National Action Plan on Climate Change (NAPCC) emphasises **inclusive adaptation**, but requires stronger localisation at city level.
2. **Strengthening decentralised governance:** Empowering ULBs with finances and functions for context-specific planning.
   Eg: The **Fifteenth Finance Commission (2021-26)** recommended **grants for urban local bodies tied to performance and service delivery improvements**.
3. **Climate-resilient and inclusive infrastructure development:** Focus on housing, drainage, and basic services in vulnerable areas.
   Eg: AMRUT Mission aims to improve **urban infrastructure like water supply and drainage**, though inclusion gaps remain (Source: MoHUA reports).
4. **Participatory urban planning and community engagement:** Incorporating voices of marginalised communities in decision-making.
   Eg: Kerala's decentralised planning model has demonstrated success in **community participation in local development planning**.
5. **Legal and rights-based approach to urban housing:** Recognising housing as a right linked to dignity under **Article 21 (Right to Life)**.
   Eg: In **Olga Tellis vs Bombay Municipal Corporation (1985)**, the Supreme Court recognised **right to livelihood as integral to right to life**, highlighting the need for humane urban policies.

Conclusion

Bridging the gap between planning and justice requires embedding equity within climate resilience frameworks. Only a people-centric, decentralised, and inclusive approach can make Indian cities truly sustainable in the face of climate change.

Effects of globalization on Indian society.

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Q. Assess the major social risks associated with increased digital exposure among children. Examine how these risks affect societal cohesion. (10 M)

Introduction

The rapid penetration of digital platforms into childhood has fundamentally altered socialisation patterns. While offering opportunities, unchecked exposure has created multidimensional social risks affecting both individuals and collective social fabric.

Body

Major social risks associated with increased digital exposure among children

1. **Mental health vulnerabilities and behavioural changes:** Prolonged exposure leads to anxiety, depression and reduced attention span among children.
   Eg: The **Economic Survey 2023-24** flagged **social media addiction** among youth as a growing public health concern linked to **sleep disorders and anxiety**, highlighting rising dependence patterns.
2. **Cyberbullying and online harassment:** Digital anonymity facilitates bullying, leading to psychological distress and social withdrawal.
   Eg: The **NCRB Crime in India Report 2022** recorded rising cases under **cyber harassment of minors**, indicating increased exposure of children to hostile online environments.
3. **Exposure to inappropriate and harmful content:** Children encounter sexualised, violent and misleading content affecting value formation.
   Eg: Concerns raised by **Parliamentary Standing Committee on IT (2021-22)** on **unregulated OTT and social media content** impacting minors' behavioural patterns.
4. **Erosion of family and peer interaction:** Excessive screen time reduces real-world social engagement, weakening primary social institutions.
   Eg: NCERT surveys on student well-being indicate increasing **screen dependency** leading to reduced face-to-face interaction and social isolation among adolescents.
5. **Digital addiction and reduced productivity:** Algorithm-driven engagement fosters compulsive usage patterns affecting academic and social functioning.
   Eg: WHO classification of **gaming disorder (ICD-11, 2019)** highlights addictive digital behaviour impacting daily functioning among youth.

Impact of these risks on societal cohesion

1. **Weakening of social bonds and trust:** Reduced interpersonal interaction undermines empathy and collective belonging.
   Eg: Decline in **community participation and youth engagement activities** reported in multiple **NITI Aayog discussions on social capital**, indicating weakening social cohesion.
2. **Normalization of deviant and aggressive behaviour:** Continuous exposure to toxic content reshapes acceptable social norms.
   Eg: Rising cases of **online hate speech and trolling among youth communities** reflect internalisation of aggressive digital cultures.
3. **Deepening generational divide:** Divergence between digitally immersed youth and traditional value systems creates social friction.

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Eg: Increasing parental concerns noted in **CBSE advisories (2022 onwards)** on regulating screen time to bridge value gaps within families.

4. **Fragmentation of shared realities:** Algorithm-driven content creates echo chambers, limiting shared social experiences.
   Eg: Studies cited in **Economic Survey 2023-24** highlight how **algorithmic feeds** reinforce selective exposure, affecting common societal narratives.
5. **Increased vulnerability to misinformation:** Children become susceptible to fake news, affecting informed citizenship and social harmony.
   Eg: Government initiatives like **PIB Fact Check Unit (2019)** underline rising concerns over **misinformation spread among digital users**, including youth.

Conclusion

Unchecked digital exposure risks fragmenting the social fabric by weakening trust, empathy and shared values. A balanced approach combining awareness, regulation and responsible platform design is essential to safeguard societal cohesion.

Q. Risk factors of cardio-metabolic diseases are socially embedded rather than individually determined. Evaluate the role of urbanisation and lifestyle transitions. (15 M)

Introduction

Health risks in contemporary India are increasingly structured by social contexts rather than isolated personal choices. The surge in cardio-metabolic diseases reflects how urbanisation and lifestyle transitions embed risk within everyday living conditions.

Body

Social embeddedness of risk factors

1. **Built environment constraints:** Urban spaces often lack walkability, green areas and safe public infrastructure, limiting opportunities for physical activity.
   Eg: In **Delhi and Bengaluru**, traffic congestion and inadequate pedestrian infrastructure reduce routine movement (Source: **WHO Global Status Report on Physical Activity, 2022**).
2. **Food environment shaping choices:** Availability and affordability of calorie-dense processed foods influence dietary behaviour beyond individual control.
   Eg: Increased consumption of **ultra-processed foods in urban India** is linked with rising obesity (Source: **ICMR-NIN Dietary Guidelines, 2023**).
3. **Socio-economic stratification:** Health risks are mediated by income, education and access to healthcare, making vulnerability socially determined.
   Eg: NFHS-5 (2019-21) shows higher prevalence of **undiagnosed hypertension** among urban poor populations.
4. **Workplace and stress structures:** Modern work culture induces chronic stress and sedentary behaviour, contributing to metabolic disorders.
   Eg: High stress levels among **urban IT professionals** are associated with hypertension and lifestyle diseases.

Role of urbanisation

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1. **Unplanned urban growth:** Rapid expansion without adequate infrastructure increases exposure to unhealthy environments and pollution.
   Eg: High **air pollution in NCR** contributes to cardiovascular diseases (Source: **Lancet Commission on Pollution and Health, 2020**).
2. **Shift in occupational patterns:** Transition from agrarian to service-sector jobs reduces physical labour and increases sedentary lifestyles.
   Eg: Urban economies dominated by **desk-based jobs** have significantly reduced daily energy expenditure.
3. **Urban poverty and living conditions:** Informal settlements face overcrowding, poor sanitation and limited healthcare access.
   Eg: Studies in **Mumbai slums** indicate high levels of undiagnosed diabetes and hypertension (Source: **ICMR studies**).

Role of lifestyle transitions

1. **Dietary transition:** Movement from traditional diets to high-fat, high-sugar foods increases metabolic risks.
   Eg: Rising **fast food consumption among urban youth** correlates with obesity trends.
2. **Declining physical activity:** Technology-driven lifestyles reduce active engagement and increase sedentary behaviour.
   Eg: Post-**COVID-19 remote work patterns** have reduced mobility and daily exercise.
3. **Substance use patterns:** Tobacco and alcohol consumption persist due to social norms and accessibility.
   Eg: Continued tobacco prevalence despite **COTPA, 2003** (Source: **GATS-2, 2016-17**).
4. **Sleep and mental health disruption:** Irregular lifestyles and stress affect biological rhythms and metabolic functioning.
   Eg: Urban populations show higher incidence of **sleep disorders linked to diabetes and hypertension** (Source: **ICMR findings**).

Way forward

1. **Healthy urban planning:** Integrate walkability, green spaces and active transport into city design.
   Eg: Smart Cities Mission initiatives promoting public spaces and non-motorised transport.
2. **Regulating food systems:** Promote healthy diets through labelling and restrictions on unhealthy food marketing.
   Eg: FSSAI Eat Right India campaign encouraging balanced nutrition.
3. **Strengthening primary healthcare:** Enhance early screening and management of CMDs.
   Eg: NPCDCS programme focusing on prevention and control of chronic diseases.
4. **Multi-sectoral governance:** Align policies across health, urban development and transport sectors.
   Eg: WHO's Health in All Policies approach emphasising inter-sectoral coordination.

Conclusion

Cardio-metabolic risks today are embedded in the social fabric shaped by urbanisation and lifestyle shifts. Transforming these environments is essential to make healthy living a collective possibility rather than an individual burden.

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Social empowerment, communalism, regionalism & secularism.

Salient features of world's physical geography.

Q. What are planetary magnetic fields? Explain their formation and significance in shaping planetary systems. (15 M)

Introduction

Invisible yet powerful, planetary magnetic fields act as dynamic shields that regulate planetary environments and interactions with space. Their presence reflects deep internal processes and determines the evolutionary trajectory of planetary systems.

Body

What are planetary magnetic fields

1. **Magnetic field as a planetary force field:** A planetary magnetic field is a region around a planet where magnetic forces dominate due to internally generated electric currents, creating a magnetosphere that interacts with solar wind.
   Eg: Earth's magnetosphere, extending thousands of kilometres, deflects charged solar particles and forms the **Van Allen radiation belts**, as confirmed by **NASA missions**.
2. **Magnetosphere as a spatial envelope:** The magnetic field forms a protective cavity called the magnetosphere, shaping the boundary between planetary atmosphere and interplanetary space.
   Eg: Jupiter's magnetosphere, the largest in the solar system, extends up to **7 million km toward the Sun**, as observed by **NASA's Juno mission (2016–present)**.

Formation of planetary magnetic fields

1. **Dynamo mechanism in fluid core:** Magnetic fields are generated by the motion of electrically conducting fluids (like molten iron) in the planetary core, driven by convection and rotation.
   Eg: Earth's geodynamo, powered by convection in the liquid outer core, is studied through **seismic and geomagnetic observations (USGS data)**.
2. **Role of planetary rotation:** Rapid rotation enhances the organisation of fluid motion, strengthening and stabilising magnetic field generation.
   Eg: Jupiter's rapid rotation (~10 hours) contributes to its extremely strong magnetic field, as confirmed by **Juno spacecraft findings**.
3. **Thermal and compositional convection:** Heat loss from the core and compositional differences create convection currents necessary for sustaining the dynamo process.
   Eg: Mercury's weak magnetic field, detected by **MESSENGER mission (2011–2015)**, indicates a partially molten core sustaining limited convection.
4. **Absence of dynamo in smaller planets:** Planets lacking sufficient internal heat or fluid core motion fail to generate strong magnetic fields.
   Eg: Mars, with a cooled core, has only **crustal remnant magnetism**, as revealed by **Mars Global Surveyor mission (1997–2006)**.

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Significance in shaping planetary systems

1. **Protection from solar wind erosion:** Magnetic fields shield planetary atmospheres from being stripped away by high-energy solar particles, preserving atmospheric stability.
   Eg: Mars lost much of its atmosphere due to absence of a global magnetic field, as evidenced by **NASA's MAVEN mission (2014–present)**.
2. **Regulation of space weather interactions:** Magnetic fields control the interaction between solar wind and planetary environment, influencing auroras and radiation belts.
   Eg: Auroras on Earth and Jupiter, caused by charged particles guided by magnetic fields, observed by **NASA and ESA missions**.
3. **Influence on atmospheric retention and habitability:** A stable magnetic field helps retain essential gases, supporting long-term climate stability and potential habitability.
   Eg: Earth's sustained atmosphere and liquid water, partly due to magnetic shielding, highlighted in **IPCC AR6 scientific assessments (2021)**.
4. **Indicator of internal planetary structure:** Magnetic field characteristics provide indirect evidence about core composition, temperature, and dynamics.
   Eg: Variations in **Earth's magnetic field (geomagnetic reversals)** studied through **paleomagnetism**, indicating core dynamics (Geological Survey studies).
5. **Contribution to planetary radiation environment:** Magnetic fields trap charged particles, creating radiation belts that influence surface and orbital conditions.
   Eg: Van Allen belts, discovered in **1958 (Explorer 1 mission)**, affect satellite operations and space missions.
6. **Role in comparative planetary geography:** Differences in magnetic fields across planets help explain divergent evolutionary paths within the solar system.
   Eg: Contrast between **Earth (strong field)** and **Venus (negligible field despite similar size)**, studied by **ESA's Venus Express mission**, shows different atmospheric evolution.

Conclusion

Planetary magnetic fields are not mere physical phenomena but fundamental determinants of planetary identity and evolution. Understanding them bridges internal geodynamics with cosmic interactions, shaping the future of planetary exploration and habitability studies.

Q. Examine the causes and characteristics of Super El Niño events. Analyse their impacts on global climate and extreme weather patterns. (15 M)

Introduction

The Pacific Ocean acts as a thermal engine regulating global climate, and disruptions in this system can trigger far-reaching climatic anomalies. **Super El Niño events**, representing extreme warming phases, signify heightened instability in ocean-atmosphere interactions with global repercussions.

Body

Causes of Super El Niño events

1. **Excessive sea surface temperature anomalies:** Super El Niño is marked by **SST anomalies exceeding +2°C in central/eastern Pacific**, intensifying oceanic heat release.
   Eg: 2026 projections by WMO and ECMWF models indicate potential strong El Niño conditions with unusually high SST anomalies. (Source: **WMO Climate Update 2026**)

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2. **Pronounced weakening of trade winds:** Strong relaxation of **easterly trade winds** reduces upwelling and allows warm water to shift eastward abnormally.
   Eg: During **1997–98 Super El Niño**, significant weakening of trade winds led to large-scale redistribution of warm waters. (Source: **NOAA ENSO reports**)
3. **Enhanced ocean-atmosphere coupling:** Positive feedback between ocean warming and atmospheric circulation amplifies the event's intensity.
   Eg: Bjerknes feedback mechanism intensifies warming when weakened winds further reduce upwelling, reinforcing SST rise. (Source: **NCERT Class XI Geography**)
4. **Accumulated subsurface heat content:** Build-up of **warm water volume in the western Pacific** prior to onset leads to stronger eastward propagation.
   Eg: Studies show increased **Pacific warm water volume before strong ENSO events**, linked to extreme phases. (Source: **IPCC AR6, 2021**)
5. **Influence of climate change:** Long-term warming trends may increase the frequency or intensity of extreme ENSO events.
   Eg: IPCC AR6 highlights potential increase in extreme El Niño events under warming scenarios.

Characteristics of Super El Niño events

1. **Extreme SST anomalies:** Sustained warming above **+2°C threshold** in Niño 3.4 region distinguishes it from normal El Niño.
   Eg: 1997–98 event recorded anomalies above +2.5°C, among the strongest observed globally. (Source: **NOAA**)
2. **Eastward shift of convection zones:** Rainfall and convection shift from western Pacific to central/eastern Pacific regions.
   Eg: Heavy rainfall in **Peru and Ecuador during strong El Niño phases** due to shifted convection zones.
3. **Suppression of upwelling:** Cold nutrient-rich water upwelling along **South American coast declines sharply**, affecting marine ecosystems.
   Eg: Collapse of **Peruvian fisheries during 1997–98 event** due to reduced nutrient supply.
4. **Global atmospheric teleconnections:** Alters **Walker circulation and jet streams**, impacting distant weather systems.
   Eg: Changes in **subtropical jet stream patterns** influencing rainfall anomalies globally.
5. **High persistence and intensity:** Super El Niño events tend to be **longer-lasting and more disruptive** than regular ENSO phases.
   Eg: 1997–98 event persisted for over a year, causing prolonged climatic disruptions.

Impacts on global climate and extreme weather patterns

1. **Intensification of heatwaves:** Elevated global temperatures due to release of oceanic heat into the atmosphere.
   Eg: 2026 early heatwaves in India linked to developing El Niño conditions with above-normal temperatures.
2. **Disruption of monsoon systems:** Weakening of **Indian southwest monsoon** leading to rainfall deficits.
   Eg: Historical El Niño years like **2009 saw significant monsoon deficit in India**, impacting agriculture. (Source: **IMD data**)
3. **Extreme precipitation and floods:** Enhanced rainfall in eastern Pacific and some regions due to shifted convection.
   Eg: Flooding in South America during strong El Niño events, especially Peru.
4. **Drought conditions in some regions:** Suppressed rainfall in **Australia, Southeast Asia and parts of Africa**.
   Eg: Australian droughts during El Niño years, affecting agriculture and water availability.

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5. **Marine ecosystem disruption:** Decline in productivity due to reduced upwelling and altered ocean chemistry.
   Eg: Fishery collapse in Pacific coastal regions during strong ENSO events.
6. **Cyclone and storm variability:** Changes in frequency and intensity of tropical cyclones across ocean basins.
   Eg: Reduced Atlantic hurricanes but increased Pacific cyclone activity during El Niño phases. (Source: **NOAA Tropical Outlooks**)

Conclusion
Super El Niño events underscore the growing volatility of Earth's climate system in a warming world. Strengthening predictive capacity and adaptive strategies is crucial to mitigate their cascading global impacts.

Distribution of key natural resources across the world (including South Asia and the Indian sub-continent); factors responsible for the location of primary, secondary, and tertiary sector industries in various parts of the world (including India).

Important Geophysical phenomena such as earthquakes, Tsunami, Volcanic activity, cyclone etc., geographical features and their location-changes in critical geographical features (including water-bodies and ice-caps) and in flora and fauna and the effects of such changes.

Q. Ocean-atmosphere interactions in the Pacific exert a disproportionate influence on the Indian monsoon system. Elucidate. (15 M)

Introduction
The Indian monsoon is governed by a complex interplay of regional and global climatic processes, among which **Pacific Ocean–atmosphere coupling** plays a decisive role. Variability in this system often determines the strength, timing and distribution of monsoon rainfall over India.

Body

<u>Ocean-atmosphere interactions in the Pacific and their influence on Indian monsoon</u>

1. **ENSO-driven weakening of Walker circulation:** During **El Niño**, warming of central-eastern Pacific weakens the **Walker circulation**, reducing convection over the Indian region and weakening monsoon winds.
   Eg: The **El Niño year 2015** saw India receive **about 14% deficient rainfall**, reflecting weakened monsoon circulation (**IMD Annual Climate Summary 2015**).
2. **Shift in convection zones and subsidence over India:** Pacific warming shifts major convection eastwards, causing subsidence over the Indian subcontinent and suppressing rainfall.
   Eg: In **2023**, developing **El Niño conditions** contributed to rainfall deficits in parts of **northwest and central India** (**IMD Seasonal Outlook 2023**).

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3. **Alteration of trade winds and monsoon circulation:** ENSO modifies **easterly trade winds**, affecting moisture transport from oceans to land and weakening monsoon flow.
   Eg: Studies by **Indian Institute of Tropical Meteorology (IITM Pune)** show reduced **low-level jet strength** during strong El Niño phases.
4. **Impact on monsoon onset and duration:** El Niño often delays onset and advances withdrawal, disturbing seasonal rainfall distribution.
   Eg: The **2009 El Niño year** experienced delayed onset and severe drought conditions across India (**IMD data 2009**).
5. **Influence of ENSO-neutral and La Niña phases: La Niña** strengthens Walker circulation, enhancing convection over India and resulting in above-normal rainfall.
   Eg: The **La Niña phase (2020–2022)** contributed to **above-normal monsoon rainfall** for consecutive years (**IMD Climate Reports 2020–2022**).
6. **Modification of upper tropospheric circulation:** ENSO alters **tropical easterly jet and upper-level divergence**, impacting vertical motion and rainfall intensity.
   Eg: During **El Niño 2015**, weakening of the **Tropical Easterly Jet** reduced rainfall over peninsular India (**IITM and IMD analysis**).
7. **Impact on monsoon depressions and synoptic systems:** Pacific anomalies influence the frequency of **low-pressure systems over Bay of Bengal**, critical for rainfall distribution.
   Eg: El Niño years typically witness fewer **monsoon depressions**, leading to rainfall deficits in central India (**IMD synoptic studies**).

<u>Why Pacific influence is disproportionate compared to other factors</u>

1. **Large spatial scale of Pacific Ocean:** Being the largest ocean basin, Pacific SST anomalies significantly influence global atmospheric circulation patterns.
   Eg: NOAA ENSO monitoring identifies Pacific SST anomalies as primary drivers of global climate variability.
2. **Strong teleconnections with Indian monsoon system:** Atmospheric linkages transmit Pacific anomalies to Indian region, influencing pressure gradients and winds.
   Eg: IPCC AR6 (2021) recognises ENSO as the dominant driver of **interannual monsoon variability**.
3. **High correlation with Indian monsoon variability:** ENSO phases historically show strong statistical linkage with droughts and excess rainfall years.
   Eg: Major drought years like **2002, 2009 and 2015** coincided with **El Niño events** (**IMD long-term data**).
4. **Dominance over Indian Ocean influences:** Pacific signals often override or modulate effects of **Indian Ocean Dipole (IOD)** and regional factors.
   Eg: In **2019**, despite a **positive IOD**, ENSO influenced spatial rainfall variability across India (**IMD Monsoon Review 2019**).
5. **Control over intra-seasonal variability and rainfall distribution:** ENSO affects not only total rainfall but also its temporal spread, especially second-half variability.
   Eg: Skymet Forecast 2026 highlights expected **El Niño development**, likely causing **deficient rainfall in August–September** (**Skymet Monsoon Outlook 2026**).

Conclusion

The Pacific Ocean acts as a climatic fulcrum shaping India's monsoon behaviour through powerful

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teleconnections. Enhancing forecasting by integrating ENSO dynamics is essential to mitigate risks arising from increasing monsoon variability in a warming world.

Q. Shrinkage of lakes is not merely a loss of water spread but a disruption of regional climate regulation." Examine the statement. Highlight the role of lakes in local climatic processes. (10 M)

Introduction

Lakes act as dynamic interfaces between land, water and atmosphere, regulating energy and moisture exchanges at local scales. Their shrinkage therefore disturbs not just hydrology but also the micro-climatic equilibrium of regions.

Body

Shrinkage of lakes as disruption of regional climate regulation

1. **Loss of thermal buffering capacity:** Lakes absorb heat during the day and release it slowly, moderating temperature extremes; shrinkage leads to higher diurnal temperature variation.
   Eg: In **Kashmir Valley**, reduction in **Dal Lake's water spread (CAG Report 2023)** has been linked to increasing **summer temperature variability** in surrounding urban zones.
2. **Decline in atmospheric moisture contribution:** Lakes contribute to local humidity through evaporation, supporting cloud formation and precipitation cycles; shrinking reduces this moisture flux.
   Eg: Hokersar wetland degradation has reduced **local humidity levels**, affecting **winter precipitation patterns** in Srinagar region (Wetlands International observations).
3. **Alteration of local wind circulation:** Water bodies influence land–water breeze systems that regulate local wind flows; their reduction weakens such circulations.
   Eg: Shrinking of **Wular Lake** has impacted **local breeze systems**, affecting dispersion of pollutants in North Kashmir.
4. **Reduction in carbon sequestration and cooling effects:** Lake ecosystems act as carbon sinks and regulate surface albedo; their loss increases heat absorption and contributes to warming.
   Eg: Loss of wetland vegetation in **Kashmir lakes (CAG Audit)** has reduced **carbon sequestration capacity**, intensifying local warming trends.
5. **Increased dust and heat island effects:** Dry lake beds become sources of dust and enhance heat retention, altering micro-climatic conditions.
   Eg: Desiccated patches around **Manasbal Lake** show increased **dust emissions and localized heat island effects** during summer months.

Role of lakes in local climatic processes

1. **Temperature regulation through high heat capacity:** Lakes stabilise surrounding temperatures by absorbing and releasing heat gradually.
   Eg: Presence of **large lakes like Wular** historically moderated **extreme winter cold** in nearby settlements.
2. **Evaporation-driven precipitation cycles:** Lakes contribute to evapotranspiration, aiding cloud formation and sustaining local rainfall regimes.

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Eg: Wetlands in **Kashmir basin** support **orographic precipitation cycles**, interacting with **Western Disturbances**.

3. **Humidity maintenance and fog formation:** Lakes maintain ambient humidity, influencing fog and dew formation crucial for local ecosystems.
   Eg: Dal and Nigeen lakes contribute to **winter fog formation**, supporting horticulture in Srinagar outskirts.
4. **Regulation of local wind systems:** Differential heating between land and water generates breezes that influence micro-climatic circulation.
   Eg: Lake-land breeze systems around **Wular Lake** historically influenced **local wind patterns** and air quality.
5. **Support to cryospheric and hydrological linkages:** In mountainous regions, lakes integrate glacial meltwater with climatic processes, stabilising hydrological cycles.
   Eg: High-altitude lakes in **Himalayan regions** act as buffers for **glacial melt variability**, as noted by **IPCC AR6 (2021)**.

Conclusion

The shrinking of lakes represents a silent climatic disruption, weakening natural regulatory mechanisms. Sustaining lake ecosystems is therefore essential not only for ecology but also for maintaining regional climate stability in vulnerable landscapes.

Q. Discuss the causes responsible for the increasing length and intensity of summers in midlatitudes. Assess their impact on climatic processes. Evaluate the implications for human systems. (15 M)

Introduction

Midlatitude climates are undergoing a structural transformation where summers are becoming longer, hotter and more abrupt in transition. This reflects a shift in seasonal dynamics driven by changes in Earth's energy balance and atmospheric processes.

Body

Causes responsible for increasing length and intensity of summers

1. **Enhanced greenhouse forcing:** Rising greenhouse gas concentrations increase radiative forcing, leading to sustained heat retention and prolonged summers.
   Eg: IPCC Sixth Assessment Report (2021–23) highlights rising **CO_2 levels beyond 420 ppm (NOAA, 2023)** contributing to persistent warming trends.
2. **Shift in seasonal onset and withdrawal:** Earlier onset of summer and delayed withdrawal extend the duration of summer conditions.
   Eg: Studies in **Environmental Research Letters (2023)** show **midlatitude summers increasing by ~6 days per decade**, leading to ~30 days longer summers than the 1960s.
3. **Land–ocean thermal contrast changes:** Differential warming of land and oceans alters heat storage and seasonal temperature cycles.

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Eg: Recent research shows **ocean summer duration increasing faster than land** due to lower seasonal variability.

4. **Decline in snow cover and albedo effect:** Reduced snow cover lowers surface reflectivity, increasing heat absorption and prolonging warm conditions.
   Eg: Arctic snow cover decline (NSIDC data) has accelerated warming in adjoining midlatitude regions.
5. **Rise in accumulated heat:** Non-linear increase in cumulative heat stress intensifies and prolongs summer conditions.
   Eg: Studies report **300% increase in accumulated heat since 1990** compared to the baseline period (Environmental Research Letters).

Impact on climatic processes

1. **Intensification of hydrological cycle:** Higher temperatures increase evaporation and reduce soil moisture, intensifying droughts.
   Eg: European drought (2022) caused **Rhine river water levels to drop significantly**, disrupting transport (European Environment Agency).
2. **Increase in heatwave frequency and persistence:** Longer summers favour prolonged high-pressure systems and extreme heat events.
   Eg: North American heat dome (2021) led to record temperatures in **Canada and the USA**.
3. **Disruption of precipitation regimes:** Changing thermal gradients alter rainfall patterns and seasonal precipitation cycles.
   Eg: IMD reports (2023–25) indicate **erratic monsoon onset and uneven rainfall distribution in India**.
4. **Accelerated cryospheric processes:** Early and intense summers increase glacier melt and snowmelt rates.
   Eg: ICIMOD reports highlight **rapid Himalayan glacier retreat**, increasing risks of floods.
5. **Enhanced atmospheric instability:** Increased heat energy fuels extreme events such as wildfires and storms.
   Eg: Australian bushfires (2019–20) were intensified due to prolonged heat and dryness.

Implications for human systems

1. **Rising energy demand for cooling:** Prolonged summers increase electricity consumption and strain energy infrastructure.
   Eg: India's peak electricity demand crossed 250 GW in 2023 (CEA) due to increased cooling needs.
2. **Health risks and reduced adaptability:** Sudden seasonal transitions reduce physiological adaptation time, increasing heat stress.
   Eg: WHO reports (2023) highlight rising **heat-related mortality in Europe and Asia** during prolonged heatwaves.
3. **Agricultural stress and productivity decline:** Altered growing seasons affect crop cycles and yields.
   Eg: Wheat yield decline in North India (2022 heatwave) due to early summer onset (ICAR observations).

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4. **Water scarcity and urban stress:** Increased evaporation and demand strain water resources, especially in urban areas.
   Eg: Chennai water crisis (2019) linked to prolonged dry and hot conditions affecting reservoirs.
5. **Economic productivity loss:** Heat stress reduces labour efficiency and increases economic costs.
   Eg: ILO (2023) estimates significant **loss of working hours globally due to extreme heat exposure**.

Conclusion

The lengthening and intensification of summers signify a fundamental shift in seasonal geography with cascading impacts on climate and society. Strengthening adaptive capacity and climate-resilient planning is essential to manage this emerging challenge.

Q. Discuss the role of alpine vegetation in maintaining ecological stability in the Himalayas. Analyse how recent climatic changes are altering this role. Evaluate the consequences for mountain livelihoods. (15 M)

Introduction

The **alpine ecosystems of the Himalayas**, located above ~4,000 m, represent one of the most fragile yet functionally critical ecological zones. Despite sparse vegetation, these regions regulate **water cycles, soil stability and biodiversity**, making them vital for both mountain ecology and downstream human systems.

Body

<u>Role of alpine vegetation in maintaining ecological stability</u>

1. **Hydrological regulation and snow retention:** Alpine vegetation traps snow and regulates meltwater, ensuring gradual release into river systems.
   Eg: Studies by **University of Exeter (2026, Ecography)** highlight that alpine shrubs **trap snow and delay melt**, stabilising seasonal flows in Himalayan catchments.
2. **Soil stabilisation and erosion control:** Root systems of shrubs and grasses bind thin mountain soils, preventing landslides and erosion in steep terrains.
   Eg: In **Ladakh and Spiti regions**, alpine vegetation reduces **soil erosion in permafrost-influenced slopes**, limiting sediment load in rivers.
3. **Microclimate regulation:** Vegetation modifies surface temperature and moisture conditions, creating localised microhabitats.
   Eg: Alpine plants provide **soil shading and reduce evapotranspiration**, maintaining moisture in otherwise arid high-altitude zones (IPCC Sixth Assessment Report, 2021).
4. **Biodiversity support and ecological niches:** Alpine vegetation supports specialised flora and fauna adapted to extreme conditions.
   Eg: The **Himalayan blue sheep and alpine pollinators** depend on seasonal vegetation cycles in high-altitude ecosystems.
5. **Nutrient cycling in fragile ecosystems:** Decomposition of plant matter contributes to limited nutrient availability in alpine soils.
   Eg: In **Bhutan Himalayas**, alpine shrubs enhance **soil organic carbon**, improving ecosystem productivity in nutrient-poor terrains.

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Impact of recent climatic changes on this role

1. **Upward shift of vegetation zones:** Rising temperatures and reduced snow depth are pushing vegetation to higher altitudes, altering ecological boundaries.
   Eg: A **2026 Ecography study** shows vegetation shifting upward by up to **6.95 m/year in Manthang (Nepal)**, indicating rapid ecological transition.
2. **Declining snow cover affecting water regulation:** Reduced snow depth weakens the ability of vegetation to regulate meltwater timing.
   Eg: Observed **decline in snow cover across western Himalayas** disrupts seasonal water flows, affecting river regimes.
3. **Emergence of greening and browning trends:** Climate variability is causing uneven vegetation responses, with some areas gaining and others losing biomass.
   Eg: Khumbu and Bhutan regions show increased **browning trends linked to precipitation variability** (University of Exeter study, 2026).
4. **Alteration of species composition:** Warmer conditions favour woody shrubs over grasses, changing ecosystem structure.
   Eg: Expansion of **woody shrubs in alpine zones** alters grazing patterns and reduces herbaceous plant diversity.
5. **Increased ecosystem instability:** Rapid climatic shifts reduce resilience of alpine ecosystems to disturbances.
   Eg: IPCC AR6 notes that **mountain ecosystems are highly sensitive**, with accelerated warming exceeding global averages.

Consequences for mountain livelihoods

1. **Impact on pastoral livelihoods:** Changes in vegetation composition affect availability and quality of grazing resources.
   Eg: In **Trans-Himalayan regions**, pastoralists face reduced **fodder availability due to shrub encroachment**, affecting livestock productivity.
2. **Water security challenges:** Altered hydrological cycles affect availability of drinking water and irrigation downstream.
   Eg: Himalayan rivers like **Indus and Ganga depend on regulated meltwater**, now increasingly variable due to ecological changes.
3. **Increased disaster vulnerability:** Loss of vegetation stability can enhance risks of landslides and soil erosion.
   Eg: Fragile slopes in **Uttarakhand Himalayas** have shown higher susceptibility to **landslides under changing vegetation cover**.
4. **Loss of ecosystem services and livelihoods diversification:** Declining ecosystem stability reduces availability of non-timber forest products and ecological services.
   Eg: Communities in **Bhutan Himalayas** rely on alpine ecosystems for **medicinal plants**, which are now shifting altitudinally.
5. **Cultural and traditional livelihood disruptions:** Indigenous practices tied to seasonal vegetation cycles are being affected.
   Eg: Traditional **transhumance systems** in Himalayan regions are disrupted due to shifting grazing zones and climatic unpredictability.

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Conclusion

The transformation of alpine vegetation reflects deeper climate-induced shifts in Himalayan ecosystems, with cascading impacts on ecology and livelihoods. Strengthening **climate-resilient mountain governance and ecosystem-based adaptation** is essential to safeguard both fragile environments and dependent communities.

Q. Discuss the role of marine heatwaves in shaping the intensity and destructiveness of tropical cyclones. Analyse their impact on coastal ecosystems. Suggest measures for climate-resilient coastal planning. (15 M)

Introduction

Oceans are increasingly acting as amplifiers of climatic extremes rather than mere heat sinks. The rise in **marine heatwaves (MHWs)** is intensifying tropical cyclones and reshaping coastal vulnerabilities.

Body

Role of marine heatwaves in shaping cyclone intensity and destructiveness

1. **Enhanced latent heat supply:** Marine heatwaves elevate sea surface temperatures, increasing the oceanic heat energy available for cyclone formation and strengthening.
   Eg: A 2026 study in **Science Advances** found that cyclones over MHWs record **up to 20% higher wind speeds**, indicating direct energy transfer from warmer oceans.
2. **Acceleration of rapid intensification:** MHWs create favourable thermodynamic conditions that enable cyclones to intensify rapidly within short durations.
   Eg: Analysis of **1600 cyclones (1981–2023)** shows higher frequency of **rapid intensification events** when storms pass over anomalously warm waters.
3. **Increased precipitation intensity:** Warmer oceans enhance atmospheric moisture content, resulting in extreme rainfall during cyclones.
   Eg: MHW-associated cyclones show **up to 12% higher precipitation rates**, aggravating flood risks during landfall.
4. **Amplified economic losses:** Intensified cyclones driven by MHWs result in disproportionately higher damages compared to exposure alone.
   Eg: Studies indicate **60% more billion-dollar disasters** linked to cyclones influenced by marine heatwaves.
5. **Slower movement and persistence:** Warmer ocean conditions contribute to slower-moving cyclones, increasing duration of impact over land.
   Eg: Events like **Hurricane Helene (2024)** stalled over land, causing prolonged flooding due to sustained energy from warm waters.

Impact on coastal ecosystems

1. **Coral bleaching and habitat loss:** Elevated sea temperatures during MHWs stress coral ecosystems, reducing their resilience to cyclone disturbances.
   Eg: Repeated bleaching events in the **Great Barrier Reef (recent decades)** have weakened reef structures, limiting their protective function against storm surges.

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2. **Mangrove degradation:** Cyclone intensity combined with thermal stress affects mangrove health and regeneration capacity.
   Eg: In the **Sundarbans**, stronger cyclones like **Amphan (2020)** caused large-scale mangrove damage, reducing coastal buffering capacity.
3. **Salinisation of coastal ecosystems:** Intense storm surges push saline water inland, altering soil and freshwater ecosystems.
   Eg: Post-cyclone **Aila (2009)**, agricultural lands in coastal West Bengal faced prolonged **salinity intrusion**, affecting biodiversity.
4. **Disruption of marine biodiversity:** Changes in ocean temperature alter species distribution and breeding cycles.
   Eg: Marine heatwaves in the **Indian Ocean** have led to shifts in fish migration patterns, impacting local fisheries.
5. **Loss of natural coastal buffers:** Damage to reefs, mangroves and wetlands reduces natural defence systems against cyclones.
   Eg: Studies by **UNEP** highlight that degraded coastal ecosystems significantly increase vulnerability to storm surges.

Measures for climate-resilient coastal planning

1. **Integration of MHW monitoring in forecasting systems:** Incorporating sea surface temperature anomalies into cyclone prediction models improves early warning.
   Eg: Advanced models combining cyclone and MHW data have shown higher accuracy in predicting **rapid intensification events**.
2. **Ecosystem-based coastal protection:** Restoration of mangroves, coral reefs and wetlands enhances natural resilience.
   Eg: India's Coastal Regulation Zone (CRZ) framework promotes mangrove conservation as a buffer against cyclones.
3. **Climate-resilient infrastructure development:** Designing infrastructure to withstand higher intensity cyclones reduces disaster losses.
   Eg: Post **Odisha Super Cyclone (1999)**, cyclone shelters and resilient housing reduced casualties in later events like **Cyclone Fani (2019)**.
4. **Land-use regulation and zoning:** Restricting development in high-risk coastal zones minimises exposure to intensified cyclones.
   Eg: Implementation of **CRZ norms** helps regulate construction in ecologically sensitive coastal areas.
5. **Strengthening institutional frameworks:** Coordinated governance and disaster preparedness improve adaptive capacity.
   Eg: Recommendations of the **NDMA guidelines on cyclone management** emphasise early warning systems and community preparedness.

Conclusion

Marine heatwaves are redefining cyclone risks by intensifying both physical and socio-economic impacts. Integrating oceanic dynamics with ecosystem-based planning is essential for building resilient and adaptive coastal systems.

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Q. Explain the concept of humid heat and its distinction from dry heat. Discuss why it poses a greater risk to human survival in tropical climates. (10 M)

Introduction

Heat stress is increasingly being shaped not just by temperature but by the body's ability to dissipate heat. In tropical climates, the interaction of heat and humidity is emerging as a critical determinant of human survivability.

Body

Concept of humid heat and distinction from dry heat

1. **Humid heat as combined thermal stress:** Humid heat refers to conditions where **high temperature and high humidity** coexist, reducing the efficiency of evaporative cooling from the human body.
   Eg: In **coastal Kerala (2024–25 heat episodes)**, people reported severe heat stress even at **moderate temperatures**, indicating humidity-driven discomfort. (Source: **Climate Dynamics study, 2026**)
2. **Role of wet-bulb temperature:** Humid heat is best measured through **wet-bulb temperature**, which integrates both temperature and humidity and reflects physiological heat limits.
   Eg: Studies show that **wet-bulb temperatures approaching 35°C** can be fatal even for healthy individuals under prolonged exposure. (Source: **Climate science literature cited in Climate Dynamics, 2026**)
3. **Dry heat and efficient cooling:** In dry heat conditions, **low humidity allows sweat to evaporate efficiently**, enabling effective body cooling despite high temperatures.
   Eg: In **northwest India (Rajasthan)**, despite temperatures exceeding **45°C**, lower humidity allows better heat tolerance compared to humid regions.
4. **Perceptual vs physiological difference:** Dry heat is often more perceptible due to extreme temperatures, whereas humid heat is **less visible but physiologically more dangerous**.
   Eg: Reports of **heat-related illnesses in Kerala without official heatwave declaration** highlight this invisible risk. (Source: **State disaster management observations, recent years**)

Why humid heat poses greater risk to human survival in tropical climates

1. **Failure of evaporative cooling mechanism:** High humidity prevents sweat evaporation, causing **rapid rise in core body temperature**, leading to heatstroke.
   Eg: During **humid heat events in coastal India**, cases of **heat exhaustion and dehydration** rise even during early morning work hours. (Source: **CEEW assessments, recent studies**)
2. **Elevated night-time temperatures:** Humid regions experience **warmer nights**, reducing the body's recovery time from daytime heat stress.
   Eg: In **peninsular India**, rising **night-time temperatures over the last decade** have increased cumulative thermal stress. (Source: **IMD climate trends**)
3. **Monsoon-linked persistence of moisture:** Even during monsoon break phases, **residual atmospheric moisture** combined with rising temperatures sustains humid heat conditions.
   Eg: Research shows that **monsoon intraseasonal variability** shifts humid heat risk across regions. (Source: **Climate Dynamics, 2026**)

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4. **Impact on labour productivity:** Tropical economies dependent on outdoor labour face **reduced working hours and productivity losses** due to heat stress.
   Eg: Construction and fishing communities in Kerala report **early fatigue and reduced work efficiency** under humid heat conditions.
5. **Higher vulnerability of populations:** Humid heat disproportionately affects **low-income groups, elderly, and those without access to cooling infrastructure**.
   Eg: Urban informal settlements in coastal cities show higher incidence of heat stress due to **poor ventilation and high density**.

Conclusion

Humid heat represents a shift from visible to physiological climate risks, demanding a rethinking of heat assessment frameworks. Integrating humidity-sensitive indices into planning is essential for safeguarding human survival in tropical regions.

Q. "Anti-cyclonic circulations over adjacent seas can significantly alter inland precipitation regimes." Analyse the mechanisms involved. Evaluate their role in recent pre-monsoon rainfall patterns. (10 M)

Introduction

Pre-monsoon weather over India is no longer a simple transition phase but a complex interplay of pressure systems and moisture dynamics. The growing influence of oceanic circulations highlights the interconnectedness of sea–atmosphere processes in shaping inland rainfall variability.

Body

<u>Anti-cyclonic circulations and alteration of inland precipitation regimes</u>

1. **Moisture transport through peripheral winds:** Anti-cyclones induce clockwise circulation, directing moisture-laden winds from seas towards land, enhancing inland humidity.
   Eg: In **March 2026**, anti-cyclones over the **Bay of Bengal** transported moisture into eastern India, leading to **excess rainfall in West Bengal (231%) and Bihar (215%)** (Source: **IMD data analysis**).
2. **Onshore wind convergence and instability:** These systems facilitate convergence with local heating-induced winds, triggering convective activity and thunderstorms.
   Eg: Increased **thunderstorm frequency across Northeast India** during pre-monsoon 2026 due to interaction of **moist onshore winds and high surface heating** (Source: **IMD observations**).
3. **Moisture feeding into synoptic systems:** Anti-cyclones act as moisture reservoirs, supplying water vapour to western disturbances and troughs, intensifying rainfall.
   Eg: Western disturbances in **March 2026** drew moisture from the **Arabian Sea anti-cyclone**, resulting in **widespread rainfall across Northwest and Central India**.
4. **Enhanced atmospheric moisture holding capacity:** Warm conditions increase the capacity of air to hold moisture, making anti-cyclone-driven supply more effective in generating precipitation.
   Eg: Temperature anomalies of 3–5°C above normal in early March 2026 amplified rainfall intensity when moisture was available (Source: **IMD, K J Ramesh analysis**).

<u>Evaluation of their role in recent pre-monsoon rainfall patterns</u>

1. **Shift towards anomalous excess rainfall:** Anti-cyclones contributed to unusually high rainfall in regions typically dry during pre-monsoon months.

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Eg: Nagaland (131%) and Assam (129%) excess rainfall in March 2026, reversing previous seasonal deficits (Source: **IMD data**).

2. **Expansion of rainfall spatial coverage:** Their presence enabled rainfall across diverse regions including eastern, southern and even western India.
   Eg: Karnataka (136%) and Gujarat receiving heavy rainfall during pre-monsoon 2026 indicates **pan-India spread**.
3. **Intensification of extreme weather events:** Moisture availability led to frequent thunderstorms, hailstorms and heavy rain episodes.
   Eg: Around **20 days of heavy rainfall events in March 2026**, accompanied by **hailstorms**, showing increased extremes.
4. **Increased interaction with multiple weather systems:** Anti-cyclones enhanced the impact of concurrent systems like troughs and cyclonic circulations, creating compound effects.
   Eg: Presence of **5–13 simultaneous weather systems daily in March 2026** amplified rainfall variability (Source: **IMD analysis**).

Conclusion

Anti-cyclonic circulations have emerged as critical modulators of India's pre-monsoon climate, reshaping rainfall regimes beyond traditional patterns. Understanding their evolving dynamics is essential for improving weather forecasting and climate resilience strategies.

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